Detecting Lookahead Bias in LLM Forecasts: A Recall-Based Diagnostic

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Abstract

We develop a statistical procedure to detect lookahead bias in economic forecasts generated by large language models (LLMs). Using a date-only recall query for a firm-date pair, we estimate the probability that the LLM has internalized information about the realized outcome, a statistic we term Lookahead Propensity (LAP). LAP is materially positive throughout the in-sample period and collapses essentially to zero right after the training-data cutoff. We show that a positive interaction between LAP and the LLM forecast in an accuracy regression indicates lookahead-bias contamination, and apply the test to two forecasting tasks: news headlines predicting stock returns and earnings-call transcripts predicting capital expenditures. In both applications, the LLM forecast's predictive power is amplified on high-LAP firm-date pairs, and the interaction loses significance on post-training-cutoff samples. Two boundary cases, CPI inflation and Chinese stocks, confirm the diagnostic's behavior at the extremes. The findings replicate across model families. Our test provides a portable, cost-efficient diagnostic tool for assessing the validity and reliability of LLM-generated forecasts.

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1 Introduction

Large language models (LLMs) are increasingly used as measurement instruments in empirical economics and finance, extracting signals from news headlines, earnings-call transcripts, and other unstructured text to predict stock returns, corporate investment, and macroeconomic outcomes (e.g., Lopez-Lira and Tang, 2026; Jha, Qian, Weber, and Yang, 2024). A growing body of work treats these LLM-extracted signals as valid, real-time measures of expectations, sentiment, or economic conditions. But how reliable are these measures? When a pre-trained model is evaluated on dates inside its training corpus, the realized outcome may itself have been ingested during pretraining. The resulting predictability blends two indistinguishable mechanisms: genuine economic reasoning and recall of memorized post-event information (Sarkar and Vafa, 2024). Any analysis of a historical episode with a modern LLM falls inside the model's training data, and the contamination frontier advances with each new model generation.[1]

This paper shows that a substantial share of the apparent predictive content of LLM-based forecasts in finance reflects memorization of training-time material rather than reasoning over the supplied input. In the headline application, where an LLM classifies news sentiment to predict next-day stock returns, a one-standard-deviation increase in the model's memorization strength raises its forecast's marginal predictive power by about 32%; in the earnings-call application, where an LLM reads a transcript to predict capital expenditure, the analogous boost is about 13%. On the subset of firm-dates where the model shows no detectable directional recall, the headline signal loses all predictive power, while the earnings signal retains 94% of its. The contamination is task-specific, substantial in two major application domains, and varies with the diagnostic's operating range. We emphasize that the diagnostic provides a necessary condition, not a sufficient condition, for a lookahead-bias-free forecast: passing the test (insignificant $\hat{\beta}_3$) does not certify the absence of memorization, because the recall query probes only one channel (explicit directional recall) and the test may lack power in the saturation or zero-exposure regimes. Failing the test (significant $\hat{\beta}_3$), however, is robust evidence that the forecast loads on memorized content through the directional-recall channel.

[1] Our diagnostic applies to prediction tasks where the outcome is realized after the text date. For classification tasks (sentiment labeling, topic coding), where the "outcome" is the label itself, the contamination is embedded in the label assignment and has no natural out-of-sample check.

Several mitigation strategies have been proposed, including masking firm identifiers (Glasserman and Lin, 2024; Engelberg, Manela, Mullins, and Vulicevic, 2025; Wu, Yang, Ying, and Zhou, 2025), restricting evaluation to post-cutoff data, and training leak-free models from scratch on time-stamped corpora (Sarkar, 2024; He, Lv, Manela, and Wu, 2025; Kelly, Malamud, Schwab, and Xu, 2026). Some studies, notably Lopez-Lira and Tang (2026), evaluate their LLM forecasts entirely out of sample, sidestepping the contamination problem by design. This is the cleanest approach where feasible, but it requires that the post-cutoff window be long enough and economically representative; for many applications, including those that study historical episodes, long-horizon return predictability, or low-frequency outcomes, a purely post-cutoff analysis is either infeasible or insufficiently powered. Moreover, even when out-of-sample performance is reported, most studies also present in-sample results to document the economic magnitudes and mechanisms, and those in-sample estimates remain vulnerable to contamination. Our diagnostic complements the out-of-sample strategy by telling the researcher how much of the in-sample performance loads on memorization.

We develop such a diagnostic. Let $Y_{i,t+h}$ denote the realized outcome of interest (e.g., stock return, capital expenditure, or inflation change), and let $X_t = \{\text{Agent}_i, \text{Time}_t, \text{Text}_t\}$ denote the information set available to the forecasting prompt, where Agent_i is the entity identifier (firm name, ticker, or country), Time_t is the observation date, and Text_t is the unstructured input (headline, transcript, or other context). The forecasting prompt uses all of X_t. The key idea is to construct a separate *recall query* that uses only $\{\text{Agent}_i, \text{Time}_{t+h}\}$, the entity identifier and the *realization* date, and nothing else: no headline, no transcript, no realized outcome, no contemporaneous information of any kind. We send this query to the LLM and read off three first-token answer-position probabilities: P_{up}, P_{down}, and P_{unknown}. Our *Lookahead Propensity*, denoted $\text{LAP}^{P(\text{known})}$, is the total probability the model assigns to the two directional labels,

$$\text{LAP}^{P(\text{known})} = P_{\text{up}} + P_{\text{down}}.$$

A value near one indicates that the model commits to a direction: it "knows" or believes it knows the outcome on that date. A value near zero indicates abstention via the unknown label, signalling no recall. Because the recall query carries no contemporaneous information, any non-trivial value of $\text{LAP}^{P(\text{known})}$ most plausibly reflects associations absorbed during training.[2]

Three regimes of the in-sample $\text{LAP}^{P(\text{known})}$ distribution determine the interpretation and power of our test, while out-of-sample $\text{LAP}^{P(\text{known})}$ should be near zero in all cases. First, when in-sample $\text{LAP}^{P(\text{known})}$ has substantial cross-sectional variation, with some firm-dates near zero and others near one, the interaction test has power to detect contamination from the within-sample covariance between forecast accuracy and memorization strength. Our headline application falls in this regime. Second, when in-sample $\text{LAP}^{P(\text{known})}$ is uniformly close to zero, meaning the model recalls essentially nothing, the interaction test is uninformative because there is no detectable directional recall, and the researcher routes to post-cutoff performance for certification. Our Chinese stock market application illustrates this case. Third, when in-sample $\text{LAP}^{P(\text{known})}$ is uniformly close to one, meaning the model has memorized everything, the interaction term collapses onto the main effect and the test loses power through collinearity (the saturation regime of Proposition 2). Our CPI inflation application illustrates this boundary. In all three cases, the out-of-sample $\text{LAP}^{P(\text{known})}$ collapses to approximately zero, confirming the verification gap.

We first verify that this construction in fact picks up training-time exposure rather than noise. Figure III plots the in-sample mean of $\text{LAP}^{P(\text{known})}$ year by year for both the stock-news application (Panel A) and the earnings-call application (Panel B). In both panels, the mean is materially positive throughout the in-sample period and then collapses essentially to zero in the year that immediately follows Llama-3.3-70B's December 2023 training-data cutoff. The model retains and often memorizes firm-day or firm-quarter outcomes from years it was trained on, and abstains on years it was never exposed to. This sharp cross-cutoff collapse is consistent with the memorization interpretation: the magnitude of $\text{LAP}^{P(\text{known})}$ tracks whether the realized outcome was available during training.

The Kodak loan announcement of July 28, 2020, illustrates the mechanism vividly. After Kodak announced a $765 million U.S. International Development Finance Corporation loan to produce ingredients for COVID-19 drugs, its stock traded up roughly 318% the following day, accompanied by multiple trading halts. The episode received intense media coverage in the months that followed and almost certainly entered the training corpus of any modern LLM. When we send Llama-3.3-70B nothing but the firm name, ticker, and date

[2] Our LAP measure differs from the Mean Percent Error (MPE) measure proposed by Lopez-Lira, Tang, and Zhu (2025), which is calculated as the difference between the LLM's estimate and the actual outcome when the model is prompted to estimate a specific economic quantity, such as the stock price on a particular day.

2020-07-29, the model assigns $P_{\text{up}} \approx 0.9999$, with the `down` and `unknown` labels each receiving probability below 10^{-5}, an essentially saturated recall on a query that contains no headline at all. A backtest that prompts the model with the original Kodak headline and treats its sentiment classification as a forecast is therefore not necessarily measuring the model's ability to reason about a corporate-financing announcement; it may be measuring the model's ability to recall what came next.

Our framework formalizes this concern. The LLM's prediction $\hat{\mu}$ is modelled as the sum of a true conditional expectation and a memorization-induced contamination term whose magnitude is the product of two structural objects: a firm-period memorization strength L_t (observable through $\text{LAP}_{i,t+1}^{P(\text{known})}$) and a model-level contamination loading γ (unobserved, identifying the model's overall propensity to apply memorized content into text-conditioned forecasts). The standard accuracy regression of the realized outcome Y_{t+1} on $\hat{\mu}$ cannot distinguish reasoning from memorization. Augmenting the regression with the LAP main effect and the interaction $\hat{\mu} \times \text{LAP}^{P(\text{known})}$ does. We prove that, under mild conditions, a positive interaction coefficient is a one-sided diagnostic for $\gamma > 0$, i.e., for the application of memorized content into the forecast. The interaction-term significance test is therefore a portable, model-level diagnostic that requires no retraining, no access to proprietary training data, and no out-of-sample window. Because $\text{LAP}^{P(\text{known})}$ is a noisy proxy for the model's true latent memorization, classical errors-in-variables bias shrinks the estimated interaction coefficient toward zero. The test is therefore conservative: a significant $\hat{\beta}_3$ is robust evidence of contamination, while an insignificant $\hat{\beta}_3$ may reflect either genuine absence of contamination or measurement-error-induced downward bias. We formalize this asymmetry and provide a minimum detectable effect calculation in Remark 1.

We apply the full detection procedure in two forecasting applications and supplement them with two boundary cases that probe the diagnostic's behavior at the extremes of the memorization distribution. The first follows Lopez-Lira and Tang (2026), who use news headlines to predict next-day stock returns over the out-of-sample period of LLMs. We replicate their exercise but use the in-sample period. We find that the LLM's headline-direction signal predicts returns; in the in-sample window, a one-step move in the LLM signal predicts a 0.21% higher next-day return. Augmenting the regression with the LAP interaction yields a positive and highly significant coefficient: a one-standard-deviation increase in $\text{LAP}^{P(\text{known})}$ raises the marginal effect of the LLM signal by 0.067 percentage points, about 32% of the standalone LLM effect. A horse race against the inner-confidence measure of Chen, Didisheim, and Somoza (2024) shows that LAP and inner confidence load on distinct components of LLM forecasting performance. Year-by-year time series of $\text{LAP}^{P(\text{known})}$ collapse sharply once the recall date crosses Llama-3.3-70B's December 2023 training cutoff, and $\text{LAP}^{P(\text{known})}$ collapses to essentially zero on the post-cutoff (year 2024) sample, exactly as the lookahead-bias hypothesis predicts.

A separate empirical exercise is also informative on its own. Because the recall query contains no headline content, the directional component $(U-D)_{i,t+1} \equiv P_{\text{up}} - P_{\text{down}}$ is by construction a pure recall signal. We show that this signal alone predicts next-day returns with a t-statistic of 3.53 in the pooled in-sample window, and the predictive content is concentrated entirely on the firm-days where $\text{LAP}^{P(\text{known})}$ is high. This validates $\text{LAP}^{P(\text{known})}$ as a sharp marker of whether the model has memorized something about the firm-date pair, independently of the headline-direction prompt.

Our second application follows Jha, Qian, Weber, and Yang (2024) and uses earnings-call transcripts to predict capital expenditures two quarters ahead. The pattern is qualitatively the same. The baseline LLM signal predicts future capex; the LAP interaction is positive and significant ($t = 2.21$); a one-standard-deviation increase in $\text{LAP}^{P(\text{known})}$ raises the marginal LLM effect by about 13% of the standalone coefficient; and $\text{LAP}^{P(\text{known})}$ collapses to essentially zero on a post-cutoff window in which the recall quarter falls strictly after the model's training-data cutoff.

The economic implication is that researchers using LLMs as measurement instruments for financial forecasting face a first-order reliability concern. The contamination is not uniform: it depends on the input domain, the target variable, the model, and the prompt. News headlines, which overlap heavily with training corpora, are severely contaminated; earnings-call transcripts, which carry genuine economic content, are moderately contaminated; CPI inflation is fully memorized but the test cannot separate memorization from skill; and Chinese stocks, which are underrepresented in the English-language training corpus, show no detectable contamination. This task-specificity means that each LLM forecasting application requires its own audit. The $\text{LAP}^{P(\text{known})}$ diagnostic we develop provides such an audit at low cost.

Related Literature

A growing body of work uses LLMs as measurement instruments in finance (e.g., Chen and Sarkar, 2020; Lopez-Lira and Tang, 2026; Jha, Qian, Weber, and Yang, 2024; Hansen and Kazinnik, 2024; Bybee, 2023; Ke, 2024). A second strand flags the possibility that part of this predictive content reflects training-time exposure (Glasserman and Lin, 2024; Sarkar and Vafa, 2024; Lopez-Lira, Tang, and Zhu, 2025; Cao, Jiang, and Xu, 2026). Mitigation strategies include anonymization (Engelberg, Manela, Mullins, and Vulicevic, 2025; Wu, Yang, Ying, and Zhou, 2025) and time-stamped models (Sarkar, 2024; He, Lv, Manela, and Wu, 2025; Yan et al., 2026; Kelly, Malamud, Schwab, and Xu, 2026). Our paper extends this literature beyond the existence question by quantifying how much of LLM forecasting performance loads on memorization, using a portable diagnostic that requires only first-token probabilities.

2 Lookahead Bias Detection

2.1 Lookahead Propensity (LAP)

We introduce the statistical properties of language models and the construction of LAP in this subsection.

Language Model A large language model (LLM) is a probabilistic framework that predicts the next token in a text sequence.[3] Given a sequence of observed tokens $w_{\leq n-1} := (w_1, w_2, \ldots, w_{n-1})$, an LLM parameterized by θ estimates the likelihood of the next token w_n through the conditional probability

$$P_\theta\big(w_n \mid w_{\leq n-1}\big) = P_\theta(w_n \mid w_1, w_2, \ldots, w_{n-1})\,.$$

LLMs learn these conditional probabilities by training on massive text corpora, adjusting internal parameters to maximize the likelihood of observed sequences.

Through repeated exposure to billions of examples, the model internalizes statistical patterns, strengthening associations between words, phrases, and events that frequently co-occur. This process does not involve "understanding" in the human sense; rather,

[3] In natural language processing (NLP), a *token* is the smallest unit of text that carries meaning, such as a word, subword, or character, depending on the tokenization scheme.

LLMs build a compressed representation of co-occurrence patterns across language. When presented with a prompt, the model retrieves relevant patterns from this learned distribution to generate contextually coherent continuations.

Critically for our analysis, this learning process absorbs not only contemporaneous associations but also retrospective narratives written after the occurrence of the event of interest, creating the potential for lookahead bias in applications that treat model outputs as real-time information.

LLM Forecast and Lookahead Bias Training data often contain both original news and subsequent articles describing market reactions, allowing models to learn events and outcomes together. In the Kodak example from Section 1, training data likely include both the loan announcement and next-day coverage of the stock surge. The model thus learns not only that "a loan announcement happened" but also that "it was followed by a sharp stock increase." This creates lookahead bias, where predictions mirror memorized outcomes rather than genuine reasoning.

When prompted with a headline consisting of N tokens $(w_1, \ldots, w_N)$, the model computes

$$P_\theta(\text{prediction} \mid \text{headline}) = P_\theta(w_{N+1} \mid w_{\leq N})$$

where w_{N+1} corresponds to the highest probability category ("positive," "negative," or "neutral").[4] These probabilities reflect statistical associations from training rather than causal reasoning. For the Kodak headline "Kodak Triples on Loan to Make Covid-19 Drug Ingredients" (Figure I), the model's prediction likely reflects memorized event-outcome pairs from its July 2020 training data.

Construction of LAP We construct $\text{LAP}^{P(\text{known})}$ from a *date-only recall query* that strips away every contemporaneous information channel and probes only the model's memorized association between a firm-date pair and the realized direction of the outcome variable. The query supplies the firm's name, ticker, and target date (or quarter), and asks the model to commit to one of three labels: `up`, `down`, or `unknown`. Because the prompt contains no headline, transcript, financial fundamentals, or any other context, any directional commitment the model produces most plausibly reflects what its training corpus encodes

[4] Since we select the maximum probability token at each step, output is deterministic given the same input.

about that firm at that point in time. A model that has memorized post-event coverage of the date will commit sharply; a model that has never seen the date should abstain.

Concretely, for each firm-day (i, t) in the stock-news application, we send Llama-3.3-70B the following prompt template, with the recall date set to the next trading day $t+1$ (illustrated for the Kodak example in Figure I, Panel B):

```
On {t+1}, did the closing stock price of {company_name}
({ticker}) go up or down compared to the previous trading day?
Answer based only on what you recall about {company_name}
({ticker}) on that specific date.  If you do not recall, answer
‘‘unknown’’.
Respond with exactly one word and nothing else:  up, down, or
unknown.
```

The placeholder `{t+1}` is the calendar date of the next trading day after t (e.g., 2020-07-29).

For each firm-quarter (i, q) in the earnings-call application, we send the analogous capex-direction query at the realization horizon $q+2$ (Figure II, Panel B):

```
In Q{q+2} {year}, did the capital expenditure of {company_name}
({ticker}) increase or decrease compared to the previous
quarter?
Answer based only on what you recall about {company_name}
({ticker}) in that specific quarter.  If you do not recall,
answer ‘‘unknown’’.
Respond with exactly one word:  up, down, or unknown.
```

We align each earnings call at quarter q with the recall query at quarter $q+2$, matching the realization horizon of $\text{CapEx}_{i,q+2}$.

The Kodak headline in Figure I illustrates a setting where memorization is starkly visible. On July 28, 2020, at the height of pandemic-driven equity volatility, Kodak announced a \$765 million U.S. International Development Finance Corporation loan to produce ingredients for COVID-19 drugs. The next trading day delivered an extraordinary +318% intraday move, accompanied by multiple trading halts; the episode received intense and prolonged coverage in financial media and almost certainly entered Llama-3.3-70B's training corpus. When we feed the model nothing but the firm name, ticker, and date 2020-07-29, it assigns $P_{\text{up}} \approx 0.9999$ to the up label, with P_{down} and P_{unknown} each below 10^{-5}, an essentially saturated recall.

A parallel pattern appears for Amazon at the onset of the pandemic (Figure II). Amazon's Q1 2020 earnings call on April 30, 2020 came amid an unprecedented capital-expenditure ramp: the firm rapidly expanded its fulfillment network, AWS data-center footprint, and logistics infrastructure to meet step-change demand growth, and the expansion was extensively reported. When we ask the model about Amazon's capex direction in Q3 2020 with no transcript or financial context, it assigns $P_{\text{up}} \approx 0.97$ to the up label, confidently recalling that capex went up. Realized capex-to-assets in fact rose from 5.5% in Q2 2020 to 9.0% in Q3 2020, a 62% jump.

Mechanically, modern LLMs are next-token predictors: at every position the model maintains a probability distribution over the next token it would emit, and we can read this distribution at the answer position before any token is sampled. Because our recall query restricts the legitimate answer to one of three tokens (`up`, `down`, or `unknown`), we simply look up the probability the model assigns to each of these three tokens at the answer position, and call them P_{up}, P_{down}, and P_{unknown}.[5] We define the *Lookahead Propensity* of firm i at day t as the total probability the model assigns to the two directional labels:

$$\text{LAP}_{i,t+1}^{P(\text{known})} := P_{\text{up}} + P_{\text{down}}. \tag{1}$$

By construction, $\text{LAP}_{i,t+1}^{P(\text{known})} \in [0,1]$. A value near one indicates that the model believes it *knows the outcome* on that date; a value near zero indicates abstention via the `unknown` label, signalling no recall. The same definition applies in the earnings-call application.

The companion directional signal is

$$(U-D)_{i,t+1} := P_{\text{up}} - P_{\text{down}}, \tag{2}$$

which indicates whether the model's recall points up or down: positive values flag an

[5] We verify empirically that these three labels capture nearly all of the answer-position probability mass: $P_{\text{up}} + P_{\text{down}} + P_{\text{unknown}}$ averages 1.00 in the headline in-sample window and 0.96 in the earnings in-sample window. The residual 4% in the earnings application falls on capitalization and spacing variants of the same labels (`Up`, ` up`, `Down`, etc.). Aggregating all case-insensitive variants into the three canonical labels raises the coverage to 0.99 and does not materially change the LAP measure. We recommend that researchers applying the procedure to a new model verify single-token coverage of their label set before proceeding.

up-leaning recall, negative values flag a down-leaning recall. The two measures are nested by construction, with $|(U-D)_{i,t+1}| \leq \mathrm{LAP}^{P(\text{known})}_{i,t+1}$, so the directional signal carries information only on firm-days where the recall itself is non-trivial.

The date-only recall query is deliberately *outcome-blind*. The prompt contains the firm's name, ticker, and target date, and nothing else: no headline, no transcript, no realized return, no fundamentals, no contemporaneous news. Whatever directional content the model produces at the answer position must therefore come from associations baked in during training.

A direct visual check of this mechanism is to plot the average $\mathrm{LAP}^{P(\text{known})}$ by year. Figure III reports the in-sample mean of $\mathrm{LAP}^{P(\text{known})}$ for the stock-news application (Panel A) and the earnings-call application (Panel B), with Llama-3.3-70B's December 2023 knowledge cutoff marked by the shaded post-cutoff region. In both panels, $\mathrm{LAP}^{P(\text{known})}$ collapses sharply once the recall date crosses the training-cutoff boundary: the model retains and often memorizes firm-day or firm-quarter outcomes from years it was trained on, but it abstains on years it was never exposed to. The contrast is precisely what a memorization story predicts. Within the in-sample window, both panels also show notable spikes in years of unusually heavy media coverage. In Panel A, mean $\mathrm{LAP}^{P(\text{known})}$ for stock news peaks at roughly 0.88 in 2020, the year of pandemic-driven equity volatility from which the Kodak example is drawn. In Panel B, mean $\mathrm{LAP}^{P(\text{known})}$ for earnings calls also peaks in 2020 at roughly 0.050, the year of the Amazon capex jump discussed earlier, and shows a second peak of roughly 0.045 in 2008, the global financial crisis. These are exactly the years when training-time exposure to firm-level outcomes should be highest under our framework.

Figures IV and V report the pooled within-window distribution of $\mathrm{LAP}^{P(\text{known})}$ for the headline and earnings applications, contrasting the in-sample window with the post-cutoff out-of-sample window side by side. For stock news, Figure IV shows a clearly bimodal in-sample distribution: roughly 42% of firm-days place essentially all mass on $P(\text{unknown})$, and roughly 23% are saturated at $\mathrm{LAP}^{P(\text{known})} \geq 0.95$. The right mass vanishes out-of-sample: across all post-cutoff firm-day recall queries in 2024, the maximum $\mathrm{LAP}^{P(\text{known})}$ is below 10^{-4} and not a single firm-day exceeds 0.5. For earnings calls, Figure V shows the in-sample distribution is sharply skewed toward zero, with the leftmost bin $[0, 0.2)$ accounting for 96.3% of mass. The small right tail above 0.5

is nontrivial: 2.1% of in-sample firm-quarters lie there, with roughly 0.6% saturated at $\mathrm{LAP}^{P(\text{known})} \geq 0.95$. It collapses cleanly out-of-sample: across all 6,619 post-cutoff firm-quarter recall queries, the maximum $\mathrm{LAP}^{P(\text{known})}$ is below 10^{-5} and every observation falls in the leftmost bin. In both applications, what disappears at the cutoff is the share of firm-period observations on which the model had confidently memorized something. For CPI inflation, the limiting case obtains: 99.8% of in-sample months are saturated at $\mathrm{LAP}^{P(\text{known})} \geq 0.9$, and every post-cutoff month is indistinguishable from zero. The model recalls the direction of essentially every historical CPI change and abstains almost completely on post-cutoff months. Together, the outcome-blind design and the sharp cross-cutoff collapse let us treat $\mathrm{LAP}^{P(\text{known})}$ as an empirical *measure* of the firm-period memorization strength $L_t \in [0,1]$ in the econometric framework of Section 2.2, with the cardinal baseline formalized below as $L_t = \mathrm{LAP}^{P(\text{known})}_{i,t+1}$. The choice of letter L is mnemonic: $\mathrm{LAP} = P(\text{up}) + P(\text{down}) \approx 1 - P(\text{unknown})$ is naturally read as the model's *likelihood* of having absorbed the firm-period outcome during training. The companion structural parameter, the model's scalar contamination loading γ, is unobserved, and identifying its sign is what the detection test of Section 2.2 delivers.

2.2 Econometric Framework

We formalize lookahead bias through a contamination model where predictions incorporate future information via memorization mechanisms. Formally, following Sarkar and Vafa (2024), lookahead bias manifests when a model's prediction $\hat{\mu}_t = \hat{\mu}(X_t)$ for time $t+1$ violates the orthogonality condition:

$$\mathrm{Cov}(\hat{\mu}_t, \varepsilon_{t+1}) \neq 0. \tag{3}$$

Consider a standard forecasting environment with the following data-generating process:

$$Y_{t+1} = \mu(X_t) + \varepsilon_{t+1}, \tag{4}$$

where Y_{t+1} denotes the realized outcome, X_t denotes the observable information at time t (e.g., news headlines and earnings-call transcripts), $\mu(X_t) = \mathbb{E}[Y_{t+1} \mid X_t]$ is the true conditional expectation, and $\varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ represents future innovations unpredictable given X_t, with $\mu(X_t) \perp \varepsilon_{t+1}$.

Definition 1 (Lookahead Bias Contamination)**.** *When a LLM suffers from lookahead bias, its predictions take the form:*

$$\hat{\mu}_t = \mu(X_t) + \gamma\, L_t\, \varepsilon_{t+1} \tag{5}$$

where $\gamma \in [0,1]$ is a model-level scalar contamination loading *(the model's average propensity to apply memorized content when producing a headline-conditioned prediction) and $L_t \in [0,1]$ is a firm-period* memorization strength *(how much of the realized outcome the model has absorbed during training for the specific firm-period $(i, t+1)$).*

The decomposition is deliberate. γ captures the model's overall propensity to deploy memorized content into its text-conditioned forecasts: a single structural parameter governing how strongly leakage flows from the recall channel into the prediction channel. L_t captures the cross-sectional/temporal heterogeneity in how much the model has memorized about firm i in period $t+1$. We measure L_t through the recall query, formalized by the working measurement assumption

$$L_t \;=\; g\big(\mathrm{LAP}_{i,t+1}^{P(\text{known})}\big), \tag{6}$$

for some monotone increasing function $g : [0,1] \to [0,1]$ with $g(0) = 0$, justified by the construction of the date-only recall query in Section 2.1: with no headline, transcript, or contemporaneous context provided, the model's $P(\text{up}) + P(\text{down})$ at the answer position is the empirical likelihood that the model has absorbed the firm-period outcome during training. Throughout the empirical analysis we adopt the cardinal baseline $g(x) = x$, so that $L_t = \mathrm{LAP}_{i,t+1}^{P(\text{known})}$. The contamination loading γ is the structural unknown and is the object identified by the detection test below.

γ as a gating parameter. A useful intuition for Equation (5) is to read γ as a *gating parameter* on the leakage channel. For each firm-period the model has memorized some amount $L_t = \mathrm{LAP}_t$ of the realized outcome (high LAP = much stored, low LAP = little or none). Whether any of this stored content flows through to the headline-conditioned forecast is governed by γ: when $\gamma > 0$, leakage is proportional to L_t (so high-LAP firm-periods carry more contamination than low-LAP firm-periods); when $\gamma = 0$, no leakage flows through at any firm-period regardless of how much has been memorized. The empirical detection test asks whether the slope of Y_{t+1} on $\hat{\mu}_t$ is amplified at high-LAP firm-periods, which, under the model's assumptions, implies $\gamma > 0$.

The key implications of our framework are as follows. First, the contamination term $\gamma L_t \varepsilon_{t+1}$ violates the orthogonality condition in Equation (3) only when both $\gamma > 0$ and $L_t > 0$ on a non-trivial set of firm-periods. Specifically, $\mathrm{Cov}(\gamma L_t \varepsilon_{t+1}, \varepsilon_{t+1} \mid L_t) = \gamma L_t \mathrm{Var}(\varepsilon_{t+1})$; unconditionally, the corresponding covariance is $\gamma \mathbb{E}[L_t \varepsilon_{t+1}^2]$, which is positive under the maintained moment conditions whenever $\gamma > 0$ and $L_t > 0$ on a set with positive probability.

Second, our framework tests two distinct empirical predictions of the contamination model through two separate regressions, corresponding to the two structural objects on either side of the gate: *existence* (does the model have non-trivial memorization?) and *application* (does that memorization transmit to the headline-conditioned forecast?).

The *validation* regression tests *existence*: that the recall channel carries outcome-relevant information, i.e., $L_t > 0$ on a non-trivial subset and the model's recall is directionally informative about the realized outcome. The directional recall signal $(U-D)_{i,t+1}$, constructed from the date-only recall query that contains no headline or other contemporaneous context, is shown to predict the realized outcome $Y_{i,t+1}$ on high-LAP firm-periods (Equations 8 and 10 in Section 3). Because the recall query is outcome-blind by construction, predictive content in $(U-D)$ on high-LAP firm-periods most plausibly originates from training-time exposure, providing evidence that the model has stored outcome-relevant content for those firm-periods.

The *detection* regression tests *application*: that the stored content actually bleeds into the headline-conditioned forecast, i.e., $\gamma > 0$. In the detection regression of Equation (7), which projects Y_{t+1} onto $\hat{\mu}_t$, LAP_t, and their interaction $(\hat{\mu}_t \times \mathrm{LAP}_t)$, the interaction coefficient β_3 is positive precisely when the LLM signal's predictive content for the realized outcome is amplified on firm-periods where stored content is high ($\mathrm{LAP}_t = L_t$ high). $\beta_3 > 0$ therefore detects $\gamma > 0$.

Third, the resulting predictor exhibits distorted accuracy. Conditional on L_t, the expected squared prediction error is

$$\mathbb{E}\left[(\hat{\mu}_t - Y_{t+1})^2 \,\middle|\, L_t\right] = \mathrm{Var}(\varepsilon_{t+1})(1-\gamma L_t)^2,$$

which falls mechanically as γL_t increases. When $\gamma = 0$ or $L_t = 0$, $\hat{\mu}_t = \mu(X_t)$ and the predictor satisfies the orthogonality condition.

Finally, two boundary cases clarify the mechanism. When $\gamma = 0$ (the model never deploys memorized content into the text-conditioned forecast) *or* $L_t = 0$ for all t (the model has memorized nothing about any firm-period), pretraining is consistent with no lookahead bias. Full contamination requires both $\gamma = 1$ and $L_t = 1$.

Under the cardinal measurement assumption $L_t = \mathrm{LAP}_{i,t+1}^{P(\mathrm{known})}$ from Equation (6), L_t is observable for every firm-period, so we estimate the detection regression directly with LAP_t in place of L_t:

$$Y_{t+1} = \beta_1 \hat{\mu}_t + \beta_2 \mathrm{LAP}_t + \beta_3 (\mathrm{LAP}_t \times \hat{\mu}_t) + \epsilon_{t+1}. \tag{7}$$

Proposition 1 (Detection Statistic)**.** *Throughout, $\hat{\beta}_3$ denotes the sample OLS interaction coefficient in Equation (7) and $\beta_3 \equiv \operatorname{plim} \hat{\beta}_3$ denotes its population (pseudo-true) counterpart. Consider the detection regression in Equation (7) under the contamination model in Equation (5) and the measurement assumption (6), and suppose the benchmark conditions stated in Appendix B.1 hold. Then, in expectation:*

(a) *Under the null $\gamma = 0$ (no contamination, equivalently $\hat{\mu}_t = \mu(X_t)$): $\beta_3 = 0$.*

(b) *Under the alternative $\gamma \in (0, 1)$, provided that $L_t = LAP_t > 0$ on a non-trivial set with positive empirical weight and that the conditional slope $\beta(a) := Cov(Y_{t+1}, \hat{\mu}_t \mid LAP_t = a)/\mathrm{Var}(\hat{\mu}_t \mid LAP_t = a)$ is non-decreasing and non-constant in a over the support of LAP_t: $\beta_3 > 0$.*

The test $\beta_3 > 0$ is therefore a one-sided test for $\gamma > 0$, i.e., for the application/transmission of memorized content into the headline-conditioned forecast, which can be verified empirically through within-bin slope estimates.

Proof: See Appendix B.1.

Interpretation and Scope Proposition 1 delivers a *qualitative*, one-sided test: it identifies the *sign* of the scalar contamination loading γ, not its magnitude. We do not point-estimate γ; the paper's claim is that $\hat{\beta}_3 > 0$ in-sample, combined with $\hat{\beta}_3 \approx 0$ out-of-sample (where $L_t = \mathrm{LAP}_t$ collapses to essentially zero), is evidence that $\gamma > 0$ on the in-sample firm-periods. Combined with the validation regression of Section 3, which provides evidence that the recall channel carries outcome-relevant information on high-LAP firm-periods (i.e., $L_t > 0$ on a non-trivial set), the two tests jointly support both the *existence* of memorization ($L_t > 0$) and its *application* into the headline-conditioned forecast ($\gamma > 0$).

Appendix B.2 makes the monotonicity condition in part (b) primitive, characterizing exactly when it holds as a function of γ, the noise-to-signal ratio, and the LAP support, and shows how to read the detection statistic when it fails.

The cardinal-scale measurement $L_t = \text{LAP}_{i,t+1}^{P(\text{known})}$ in Equation (6) can be weakened to the monotone form $L_t = g(\text{LAP}_t)$ for some monotone increasing function $g : [0,1] \rightarrow [0,1]$ with $g(0) = 0$, without affecting the substantive content of Proposition 1. Part (a) is invariant under any such reparameterization, since $\gamma = 0$ implies $\hat{\mu}_t = \mu(X_t)$ regardless of g. Part (b) carries through provided the conditional slope $\beta(a)$ remains non-decreasing on the support of LAP_t under the reparameterized $L_t = g(\text{LAP}_t)$. As a robustness check against the linear-in-LAP functional form, the detection regression can be re-estimated with LAP_t discretized into bins or replaced by its rank, which preserves the qualitative test without imposing cardinality on the LAP scale.

An important scope limitation is that the recall query probes one specific channel: explicit directional recall from a date-only prompt. Lookahead bias can also enter through channels that leave LAP near zero: qualitative narratives memorized without encoding a directional outcome, aggregate or sector-level memory (e.g., "tech was in a capex boom") that does not map to a specific firm-quarter, retrospective framing absorbed from analyst commentary or management follow-ups, and context-dependent retrieval where the full headline or transcript activates memory through cues that the stripped recall query does not. These channels are not captured by the interaction test, and the paper cannot bound how much contamination flows through them. Accordingly, the test is informative when it detects contamination: a significant $\hat{\beta}_3 > 0$ is robust evidence that memorized content enters the forecast through the directional-recall channel. It is less informative when it does not detect contamination, because the absence of directional recall does not certify the absence of other memorization channels. For the headline application, the low-LAP subsample provides indirect evidence on this question: the headline signal loses all predictive power where directional recall is absent (Section 3), suggesting that for this application, alternative channels do not carry substantial additional signal. For the earnings application, the low-LAP subsample retains 85% of the forecast's predictive content, and the paper cannot distinguish whether this reflects genuine economic reasoning in the transcript or memorization through channels that LAP does not reach.

Finally, the result clarifies why common mitigation strategies such as prompt engineering or identifier masking are ineffective: they do not eliminate the structural contamination term $\gamma L_t \varepsilon_{t+1}$ that embeds future information in LLM outputs. So long as the model has non-trivial transmission ($\gamma > 0$) and non-trivial memorization strength ($L_t > 0$) on at least some firm-periods, leakage will continue to flow.

Detection vs. prevention. Our diagnostic complements two other approaches: *prevention* through time-stamped models (He, Lv, Manela, and Wu, 2025; Yan et al., 2026; Kelly, Malamud, Schwab, and Xu, 2026) and *mitigation* through anonymization (Glasserman and Lin, 2024; Engelberg, Manela, Mullins, and Vulicevic, 2025). Prevention eliminates contamination by construction but requires retraining at smaller scale; mitigation removes identifiers but does not address memorization from non-identifying content (Wu, Yang, Ying, and Zhou, 2025). Detection via LAP tells the researcher how much of a given forecast's content loads on memorization, given the model and prompt already chosen. Appendix A provides a step-by-step implementation procedure.

3 Results

3.1 Data

Stock Market Information Daily stock returns, open prices, and close prices are obtained from the Center for Research in Security Prices (CRSP), covering all common stocks listed on the New York Stock Exchange (NYSE), the National Association of Securities Dealers Automated Quotations (NASDAQ), and the American Stock Exchange (AMEX). We restrict the sample to common stocks with a share code of 10 or 11, consistent with prior studies. We obtain financial statement variables from Compustat.

Stock News Headlines Data on news headlines are collected via web scraping from Bloomberg News. We select Bloomberg News because it is part of Bloomberg L.P., a leading global financial information provider whose real-time news service is updated more frequently than traditional outlets. We collect the news headlines for all CRSP-listed companies in the sample period and match them based on company names and ticker symbols. The in-sample dataset spans January 2012 to December 2023 and consists of 91,361 news headlines covering 1,587 unique companies. We also collect a calendar-year

2024 sample of 7,568 headlines for the post-cutoff out-of-sample placebo.

Earnings Call Transcripts We obtain data on firms' earnings call transcripts from Thomson Reuters' StreetEvents database. Then, we merge them with CRSP and Compustat data using firm tickers and corresponding call dates. The in-sample window (2006Q1–2020Q4) yields 116,746 matched firm-quarter observations from 4,225 unique U.S. publicly listed firms. We also collect 6,744 matched firm-quarters from 2023Q3 through 2024Q1 for the post-cutoff out-of-sample placebo.

3.2 LLM Setup and Prompt Design

Our analysis adopts Llama-3.3-70B, an open-source model released by Meta. Llama-3.3-70B has a publicly documented training-data knowledge cutoff of December 2023, disclosed in the official model card.[6] We adopt this as the natural pre/post split point between our in-sample window (through December 2023) and our out-of-sample window (year 2024 for stock news; 2023Q3–2024Q1 earnings calls, paired with recall queries at 2024Q1–2024Q3). The cutoff defines what the model could plausibly have memorized: anything published before December 2023 is potentially absorbed into its parameters, while anything published after is unlikely to have been seen during training.

The earnings out-of-sample window deserves explanation. Because the LAP query is aligned with the realization quarter $q+2$ (the quarter $\text{CapEx}_{i,q+2}$ materializes in), the relevant "post-cutoff" condition is that the *recall quarter* fall after December 2023, not the earnings-call quarter itself. The earliest recall quarter strictly after the cutoff is 2024Q1, which corresponds to an earnings call at $q = \text{2023Q3}$. Restricting to recall quarters in 2024 therefore yields earnings-call quarters spanning 2023Q3 through 2024Q1, the three most recent quarters available before our data cut.

Computing $\text{LAP}^{P(\text{known})}$ requires only the model's first-token probabilities at the answer position of the recall query, that is, the probabilities the model assigns to the labels up,

[6]The Llama-3.3-70B model card (`https://github.com/meta-llama/llama-models/blob/main/models/llama3_3/MODEL_CARD.md`) states a "knowledge cutoff" of December 2023. For pre-trained language models, this refers to the latest date of training data included in the pre-training corpus: the model's parameters encode statistical patterns from text published up to this date. The cutoff does not imply a sharp boundary: text published shortly before the cutoff may be incompletely represented, and web-crawled data may include documents with uncertain publication dates. We treat the stated cutoff as a conservative upper bound on the model's training exposure and verify its empirical sharpness through the year-by-year LAP collapse in Figure III.

`down`, and `unknown`. These are output-token probabilities at a single forward pass, and they are exposed by both open-source LLMs and the major closed-API providers.[7] The LAP test we develop is therefore portable across the contemporary LLM landscape.

We adopt Llama-3.3-70B in this paper for replicability. Model checkpoints can be freely downloaded from platforms such as HuggingFace, which allows any researcher to reproduce our analysis using the identical model version. In contrast, proprietary API providers may update or deprecate models over time (as OpenAI has done with ChatGPT 3.5), preventing future replication using the exact model version.

For the stock news exercise, we follow the prompt design in Lopez-Lira and Tang (2026). To ensure robustness in our analysis, we also instruct the LLM to provide a confidence score alongside its prediction. The prompt explicitly instructs the model to predict whether the news is "good", "bad", or "neutral" for the stock price of the mentioned company,[8] accompanied by a confidence score and a brief explanation, as shown in Figure I. This structured format is intended to enhance interpretability, allowing for clear numerical indicators for subsequent analysis.

For each headline, the model's response is parsed and mapped to a numerical score, where good is mapped to $+1$, neutral is mapped to 0, and bad is mapped to -1. The confidence score is directly recorded from the LLM's output, while the explanation is stored for interpretive analysis.

For the earnings call exercise, we adopt the approach of Jha, Qian, Weber, and Yang (2024), similarly incorporating confidence scores and limiting the input to the first 1,000 words of each transcript, as shown in Figure II. The prediction signal generated by the LLM for firm i in quarter q is based on its earnings call transcript. From the LLM prediction on earnings call data, the variable takes values of -1, -0.5, 0, 0.5, or 1, indicating whether future capital expenditures are expected to significantly decrease, slightly decrease, not change, slightly increase, or significantly increase before they are realized.

We use vLLM to run the Llama-3.3-70B checkpoint locally, enabling efficient batched

[7] For OpenAI, set `logprobs=true` and `top_logprobs=5` in the Chat Completions request; the per-token log-probabilities of the chosen and top alternative tokens are returned alongside the generated content. Our methodology is applicable to any open-source or closed-source model, provided we are allowed to access the logprobs for the output tokens.

[8] We also conduct robustness tests by replacing "stock price" with "stock return", as returns more directly reflect changes in market expectations. This aligns with Baker and Wurgler (2006) who showed that sentiment-driven mispricing in certain stock categories creates predictable return patterns: underpricing during low sentiment periods leads to higher subsequent returns, while overpricing during high sentiment leads to lower returns. The results remain qualitatively similar.

inference on GPU. For each recall query, we request the top-K log-probabilities at the next-token position via `logprobs = 20`, which exposes $P(\text{up})$, $P(\text{down})$, and $P(\text{unknown})$ at the answer position; $\text{LAP}^{P(\text{known})}$ is then constructed firm-by-firm as $P(\text{up}) + P(\text{down})$. We configure inference with `temperature = 0` to ensure deterministic outputs, yielding reproducible results up to minor numerical variability.[9]

3.3 Prompt News Headlines to Predict Stock Returns

In this exercise, we generally follow the approach of Lopez-Lira and Tang (2026), with a few differences. First, their study evaluates ChatGPT-4 in a post–knowledge-cutoff window, which can be interpreted as an out-of-sample setting; our main tests are conducted in the pre-cutoff (in-sample) window for Llama-3.3-70B. Second, we use news headlines from Bloomberg, while they collect headlines from multiple media sources. Third, their analysis relies on ChatGPT-4 to classify each headline as good (+1), bad (−1), or neutral (0); we adopt the same classification scheme but use Llama-3.3-70B for the reasons discussed above. As shown later, the choices on news source and model do not materially affect the baseline predictability result, as we successfully replicate their main finding.

For simplicity, we use the stock market close at 4 p.m. as the cutoff to assign headlines to days. A headline published before 4 p.m. on day t is treated as news on day t and matched to the next-day stock return $r_{i,t+1}$; a headline published after 4 p.m. is treated as news on day $t+1$.

The in-sample window spans January 2012 to December 2023 ($N = 91{,}361$ headline-day observations covering 1,587 firms). The out-of-sample window is the calendar year 2024, the period strictly after Llama-3.3-70B's December 2023 knowledge cutoff. The in-sample window is where memorization is possible; the out-of-sample window serves as a placebo where, by construction, no training-time exposure to the realized outcome could have occurred.

Validation: the recall query alone predicts returns Before testing whether LAP amplifies the LLM signal, we verify that the recall query is in fact picking up something that loads on returns. The directional signal $(U-D)_{i,t+1}$ defined in Equation (2) is

[9] We note that inference is not strictly deterministic under batched execution, due to factors such as non-deterministic GPU kernel scheduling and floating-point reduction order across parallel threads. We empirically verified that any resulting variability is negligible and does not affect our reported results.

constructed from a prompt that contains the firm name, ticker, and target date and *nothing else*: no headline, no transcript, no contemporaneous news. If the model's recall encodes information about the realized direction, that information should appear in $(U-D)$ even without any sentiment input.

To make this concrete, we estimate

$$r_{i,t+1} = \alpha_i + \lambda_t + \theta \cdot (U-D)_{i,t+1} + u_{i,t+1}, \tag{8}$$

on the pooled in-sample window and then on subsamples that split firm-days at the median of $\text{LAP}^{P(\text{known})}$. Table I reports the results. In Column (1), the pooled estimate is $\hat{\theta} = 0.264$ ($t = 3.53$): a one-unit movement in the recall-query directional signal predicts a 26 bp higher next-day return. Equivalently, a one-standard-deviation increase in $(U-D)_{i,t+1}$ predicts a 0.127 percentage points higher next-day return, about 2.95% of one return standard deviation. In Column (2), restricted to firm-days in the top half of $\text{LAP}^{P(\text{known})}$, the coefficient rises to 0.301 ($t = 3.55$). In Column (3), restricted to the bottom half, the coefficient is statistically zero (-0.190, $t = -0.37$). The recall query carries return-predictive content precisely on the firm-days where the model commits to a direction, and essentially no content where it abstains. This validates $\text{LAP}^{P(\text{known})}$ as a sharp marker of whether the model has memorized something about the firm-date pair.

Detection: the LLM signal is amplified at high LAP Our primary specification is the detection regression in Equation (7), estimated on the full in-sample window with firm and date (or quarter) fixed effects and standard errors clustered by date (or firm). All subsequent analyses in this paper, including the inner-confidence horse race, the low-LAP subsample, and the cross-model replication, are robustness checks on this primary specification.

We now turn to the main detection regression motivated by Section 2.2:

$$r_{i,t+1} = \alpha_i + \lambda_t + \varphi \cdot \text{LLM}_{i,t} + \eta \cdot \text{LAP}^{P(\text{known})}_{i,t+1} + \delta \cdot \left(\text{LLM}_{i,t} \times \text{LAP}^{P(\text{known})}_{i,t+1}\right) + u_{i,t+1}, \tag{9}$$

where $r_{i,t+1}$ is the next-day return for firm i in percentage points, $\text{LLM}_{i,t} \in \{-1, 0, +1\}$ is the headline-direction signal from Llama-3.3-70B, and $\text{LAP}^{P(\text{known})}_{i,t+1}$ is the lookahead propensity defined in Equation (1). All specifications include firm and date fixed effects, with standard errors clustered by date. Per Proposition 1, a positive interaction coefficient $\hat{\delta}$ is the empirical signature of lookahead-bias contamination.

Table II presents the results. Column (1) reports the LLM-only specification and replicates the central finding of Lopez-Lira and Tang (2026): a one-step move in the LLM signal predicts a 0.21% higher next-day return ($t = 12.18$). Column (2) adds the LAP main effect and the LLM × LAP interaction. The interaction coefficient is positive and highly significant: $\hat{\delta} = 0.162$ ($t = 3.64$). Economically, on a firm-day where $\mathrm{LAP}^{P(\mathrm{known})}$ equals one (the model is essentially certain it knows the outcome), the marginal effect of the LLM signal is $0.141 + 0.162 = 0.303\%$, more than double the marginal effect on a firm-day where $\mathrm{LAP}^{P(\mathrm{known})}$ equals zero (0.141%). Equivalently, a one-standard-deviation increase in $\mathrm{LAP}^{P(\mathrm{known})}$ raises the marginal effect of the LLM signal by 0.067 percentage points, about 32% of the standalone LLM effect reported in Column (1). The headline-direction signal performs much better on the firm-days where the model already knows the answer.

Compare to LLM Prediction Confidence Recent studies have examined how an LLM's self-reported or inferred confidence interacts with its forecast accuracy. For example, using a similar financial-news and stock-return setting, Chen, Didisheim, and Somoza (2024) show that the conditional probability of an LLM's response contains economically meaningful information and improves predictive performance. Although these confidence measures are conceptually distinct from our LAP metric, it is worthwhile to test empirically whether they capture related mechanisms.

Specifically, we incorporate two confidence measures into our analysis: (1) the first-token conditional probability in the LLM response $p(w_{i,N+1} \mid w_{i,\leq N})$, which reflects the model's inferred belief in its prediction given the wording of the prompt (Chen, Didisheim, and Somoza, 2024), and (2) the model's self-reported confidence level, $\mathrm{Confidence}_{i,t}$. We then conduct horse-race regressions between these confidence measures and LAP to assess whether model confidence and memorization propensity explain overlapping or distinct components of LLM forecasting performance.

Table III shows the results. In Column (1), we include LLM prediction, $p(w_{i,N+1} \mid w_{i,\leq N})$, and their interaction term. We find that the coefficient before the interaction term is significantly positive, consistent with findings of Chen, Didisheim, and Somoza (2024). In Column (2), we add LAP and LLM × LAP. The main finding remains robust: the coefficient on LLM × LAP is still significantly positive with a similar magnitude to that reported in Table II. The effect of $p(w_{i,N+1} \mid w_{i,\leq N})$ on LLM does not change materially either. In Columns (3) and (4), we repeat the analysis by using the model's self-reported confidence, and the results are similar.

Taken together, the result suggests that LAP and model confidence capture distinct components of LLM forecasts.

Out-of-Sample Placebo To validate that the in-sample LLM × LAP amplification is driven by memorization rather than some other source of co-movement, we re-estimate the headline regressions on the out-of-sample window. Because the lookahead-bias mechanism requires the model to have been exposed to the firm-day pair during pretraining, the interaction should disappear once we restrict the sample to firm-days that lie strictly after the model's training-data cutoff. As established in Section 3, the natural placebo window for Llama-3.3-70B is the calendar year 2024, the period strictly after its December 2023 knowledge cutoff. Figure III confirms what is required for the placebo to bite: $\text{LAP}^{P(\text{known})}$ collapses essentially to zero in 2024.

The post-cutoff collapse also disciplines an alternative interpretation of the in-sample interaction: that high-LAP firm-days are simply high-salience events where headlines are clearer and forecasts naturally more accurate, with no memorization involved. If headline salience rather than memorization drove the interaction, the amplification should persist post-cutoff, since salient events continue to occur in 2024 and the LLM's ability to read clear headlines does not degrade at the training cutoff. The fact that the LLM main effect survives post-cutoff while the LAP-dependent amplification vanishes is inconsistent with the salience interpretation: salience does not switch off at a training cutoff, but memorization does.[10]

A complementary exercise restricts the in-sample estimation to the subset of firm-days where $\text{LAP}^{P(\text{known})}$ falls below the maximum post-cutoff value (7.56×10^{-5}), i.e., the observations on which the model's recall is indistinguishable from its post-cutoff behavior. On this low-LAP subsample ($N = 4{,}574$, roughly 5% of the in-sample), the LLM coefficient

[10]We estimate the post-cutoff detection regression but do not tabulate it because $\text{SD}(\text{LAP}^{P(\text{known})})$ is on the order of 10^{-6} in the 2024 sample, rendering the interaction coefficient numerically unstable (the point estimate is inflated by the near-zero denominator). The t-statistic, which is scale-invariant, is 1.06, statistically insignificant. The LLM-only coefficient on the post-cutoff sample is 0.436 ($t = 6.47$), confirming that the headline signal retains predictive content on data the model has not seen.

is 0.063 ($t = 0.68$), statistically indistinguishable from zero. The headline-direction signal has no detectable predictive power on firm-days where the model shows no detectable directional recall of the outcome. This contrasts sharply with the full-sample estimate of 0.209 ($t = 12.18$) and the clean marginal effect at LAP $= 0$ from the interaction regression (0.141, $t = 7.55$), and suggests that a substantial share of the headline signal's apparent predictive content in the pooled sample is attributable to memorization.

3.4 Prompt Earnings Call Transcripts to Predict Capex

In our second forecast exercise, we follow the approach of Jha, Qian, Weber, and Yang (2024), who use earnings-call transcripts to predict firms' capital expenditures two quarters ahead. We adopt their classification scheme, in which Llama-3.3-70B reads a truncated transcript and assigns the firm to one of five capex-direction buckets coded as $\{-1, -0.5, 0, +0.5, +1\}$ (Section 3). The in-sample window is 2006Q1–2020Q4, identical to theirs, and our matched sample contains 116,746 firm-quarter observations covering 4,225 unique U.S. publicly listed firms.[11] The out-of-sample window consists of the most recent three earnings-call quarters available before our data cut (2023Q3 through 2024Q1), paired with recall queries at $q+2$ that fall in 2024Q1–2024Q3, i.e., strictly after Llama-3.3-70B's December 2023 knowledge cutoff.

Validation: the recall query alone predicts capex We begin with the same validation exercise as in the headline application. The directional signal $(U-D)_{i,q+2}$ is constructed from a date-only prompt that asks the model to recall whether firm i's capex went up or down at quarter $q+2$, with no transcript or financial context provided. If the model has memorized realized capex changes for firms it was trained on, that information should appear in $(U-D)$ even without any input from the transcript.

We estimate

$$\text{CapEx}_{i,q+2} = \alpha_i + \lambda_q + \theta \cdot (U-D)_{i,q+2} + u_{i,q+2}, \tag{10}$$

on the pooled in-sample window and on subsamples that split firms at the median of their

[11]The exact observation count varies slightly across tables due to variable availability and singleton fixed-effect absorption. The matched panel has 116,746 observations with non-missing capex, LLM signal, and LAP. Regressions with firm and quarter fixed effects report 116,518 after dropping singletons. Summary statistics in the Online Appendix condition on additional variables (inner confidence, salience proxies) and report correspondingly smaller samples.

firm-level time-series average of $\text{LAP}^{P(\text{known})}_{i,q+2}$. Table IV reports the results. Column (1) reports the pooled estimate $\hat{\theta} = 0.310$ ($t = 1.84$). On the half of firms with high firm-level mean LAP (Column (2)), the coefficient strengthens to $\hat{\theta} = 0.454$ ($t = 2.57$); on the low-LAP half (Column (3)), it is statistically zero ($\hat{\theta} = 103.42$, $t = 1.70$, with the large numerical magnitude reflecting near-zero variance of $(U - D)$ on this subsample). The pattern matches the headline application: the recall query carries capex-predictive content precisely on firms where the model commits to a recall, and essentially no content where it abstains.

Detection: the LLM signal is amplified at high LAP We then run the main detection regression,

$$\text{CapEx}_{i,q+2} = \alpha_i + \lambda_q + \varphi \cdot \text{LLM}_{i,q} + \eta \cdot \text{LAP}^{P(\text{known})}_{i,q+2} + \delta \cdot \left(\text{LLM}_{i,q} \times \text{LAP}^{P(\text{known})}_{i,q+2}\right) + u_{i,q+2}, \quad (11)$$

with firm and quarter fixed effects and standard errors clustered by firm. Table V reports the estimates. Column (1) reproduces the baseline finding of Jha, Qian, Weber, and Yang (2024): a one-step move in the LLM signal predicts 0.543 percentage points of higher two-quarters-ahead capex-to-assets ratio ($\text{CapEx}_{i,q+2}$, $t = 17.37$). Column (2) adds the LAP main effect and the LLM × LAP interaction. The interaction coefficient is positive and significant: $\hat{\delta} = 0.580$ ($t = 2.21$). A one-standard-deviation increase in $\text{LAP}^{P(\text{known})}$ raises the marginal effect of the LLM signal by 0.073 percentage points, about 13% of the standalone LLM effect reported in Column (1). The capex prediction is meaningfully amplified on the firm-quarters where the model already knows the realization.

The same low-LAP subsample exercise applied to the earnings application yields a different conclusion. Restricting to firm-quarters where $\text{LAP}^{P(\text{known})}$ falls below the maximum post-cutoff value (1.02×10^{-5}) yields $N = 11{,}095$ observations (roughly 10% of the in-sample). On this subsample the LLM coefficient is 0.509 ($t = 5.59$), significant and equal to approximately 94% of the full-sample estimate of 0.543 ($t = 17.37$). The earnings-call transcript carries genuine forecasting content for future capital expenditure even on firm-quarters the model shows no detectable directional recall of. The contrast with the headline application is instructive: the headline signal loses all predictive power on the low-LAP subsample, while the earnings signal retains most of its, suggesting that memorization accounts for a much larger share of the headline signal's apparent predictability than of the earnings signal's.[12]

3.5 Prompt the LLM to Predict CPI Inflation Changes

The two applications above occupy opposite ends of the memorization spectrum: stock-news headlines are bimodal (roughly half of firm-days have $\mathrm{LAP}^{P(\text{known})} \approx 0$ and a quarter are saturated near 1), while earnings calls are concentrated near zero with a thin right tail. We now apply the diagnostic to a third setting that sits at an extreme the interaction test was not designed for: macro CPI inflation, where the model has memorized essentially everything.

Following Alam, Boyle, Li, and Sekhposyan (2026), we prompt Llama-3.3-70B to forecast the direction of change in U.S. year-over-year CPI inflation at a one-month horizon, using a date-only prompt that supplies no contemporaneous macroeconomic data. The in-sample window spans January 1970 through December 2023 (647 months); the post-cutoff window covers January 2024 through July 2025 (19 months). The recall query asks: "In {YYYY-MM}, did the U.S. year-over-year CPI inflation rate go up or down compared to the previous month?"

The outcome variable is $Y_t = \Delta\pi_t^{\text{YoY}}$, the month-over-month change in U.S. year-over-year CPI inflation (in percentage points). "Up" corresponds to $Y_t > 0$ (year-over-year inflation accelerated relative to the previous month); "down" corresponds to $Y_t < 0$ (decelerated). The LLM forecast signal $\hat{\mu}_t$ is the model's predicted direction and magnitude of this change, elicited from a date-only prompt asking for the month-over-month change in percentage points.

Memorization is near-total. The contrast with the headline and earnings applications is stark. In-sample, 99.8% of months have $\mathrm{LAP}^{P(\text{known})} \geq 0.9$; the mean is 0.999. Out-of-sample, the collapse is absolute: across all 19 post-cutoff months, the maximum $\mathrm{LAP}^{P(\text{known})}$ is 0.014 and the mean is 0.001. A two-sample t-test confirms the cutoff collapse (Table A.7). The in-sample saturation holds year by year: every year from 1970 through 2023 has mean $\mathrm{LAP}^{P(\text{known})}$ at or above 0.93, and both 2024 and 2025 are indistinguishable

[12] As in the headline application, we estimate but do not tabulate the post-cutoff detection regression for earnings. With $\mathrm{SD}(\mathrm{LAP}^{P(\text{known})})$ on the order of 10^{-7}, the interaction coefficient is numerically unstable. The scale-invariant t-statistic is 1.35, insignificant. The LLM-only coefficient on the post-cutoff sample is 0.189 ($t = 2.85$), confirming genuine forecasting content in the transcript.

from zero.

CPI data is among the most widely published and discussed economic statistics; the model's training corpus contains decades of government releases, news reports, textbook discussions, and academic papers quoting specific monthly CPI figures. Near-total memorization of the historical sequence is the expected result.

The interaction test is mechanically powerless. Because $\mathrm{LAP}^{P(\mathrm{known})}$ has essentially no within-sample variation (it is concentrated at 1), the interaction regressor $\hat{\mu}_t \times \mathrm{LAP}_t$ is almost perfectly collinear with the main effect $\hat{\mu}_t$. Estimating Equation (7) on this sample, Stata omits the interaction term due to collinearity. The variance inflation factor VIF_3 is effectively infinite and the minimum detectable effect ρ_{MDE} (Remark 1) is unbounded. The interaction test cannot separate memorization from skill when there is no cross-sectional or time-series variation in the memorization measure, regardless of whether the model is using its memory or not.

This application therefore illustrates the saturation regime described in the regime-complete characterization (Proposition 2): the detection test requires variation in $\mathrm{LAP}^{P(\mathrm{known})}$ to have power, and the CPI application has none. More broadly, whenever memorization is approximately homogeneous within a period, the contamination signal is absorbed by time fixed effects and the LLM main effect, leaving no residual variation for the interaction to detect. This is a structural false negative: contamination is present but the design cannot identify it. The appropriate conclusion under our procedure is not "no contamination detected" but rather "the test is uninformative," and the researcher routes to post-cutoff performance as the binding evidence. With only 19 post-cutoff months, the power of that comparison is itself limited, which is a candid constraint of applying the diagnostic to macro time series with short post-cutoff windows.

3.6 Chinese Stock Market: CSI 300

The three applications above all use English-language inputs and U.S. economic outcomes, which are heavily represented in the training corpus of English-centric models such as Llama-3.3-70B. We now test the diagnostic on an application where the model's training-time exposure is expected to be far weaker: predicting next-day stock returns for constituents of the CSI 300, China's benchmark equity index.

We construct a daily panel of CSI 300 constituent stocks from January 2021 through December 2025 using CSMAR data. The index rebalances semi-annually; we track the constituent list at each rebalance date and match each stock-day observation to the CSMAR daily return file. We run the recall query on all stocks that were ever CSI 300 constituents during the sample period, covering all trading days. The resulting panel contains 472,932 stock-day observations covering 444 unique stocks, with 276,103 observations in the pre-cutoff window (through December 2023) and 196,829 in the post-cutoff window (2024 onward).

Near-zero exposure. We run the date-only recall query in English, asking Llama-3.3-70B whether each stock went up or down on the next trading day, using the Chinese company name and stock code. In stark contrast to the U.S. headline application, $\text{LAP}^{P(\text{known})}$ is essentially zero everywhere: the in-sample mean is 0.00004, the maximum across all 276,103 pre-cutoff observations is 0.047 (for BYD, the most internationally recognized Chinese firm), and 100% of observations fall below 0.01. Although a two-sample t-test rejects equality of in-sample and post-cutoff means (Table A.7), the difference is economically negligible: 0.00004 versus 0.000002. The nonzero mass reflects the softmax architecture of the model, which assigns positive probability to every token in its vocabulary; it does not indicate memorization. The 20-fold difference between in-sample and post-cutoff means suggests the model has absorbed a faint trace of Chinese stock outcomes through English-language sources, but the magnitude is too small to produce economically meaningful contamination.

This result is expected. Llama-3.3-70B is trained predominantly on English-language text. Chinese stock market outcomes are reported primarily in Chinese-language media, financial databases, and regulatory filings that are underrepresented in the model's training corpus. The CSI 300 application therefore provides a natural near-zero-exposure case: an application where the model's training-time exposure is negligible and the diagnostic reports no detectable directional recall. For a researcher applying our procedure to a Chinese-market LLM forecasting application using Llama-3.3-70B, the appropriate conclusion is that there is no detectable directional recall for this model and market, though the test cannot rule out memorization through channels the recall query does not reach.

3.7 Cross-Model Robustness

The detection results above use a single model, Llama-3.3-70B, for both the forecast and the recall query. We replicate the full procedure using GPT-4o-mini (OpenAI), an independently trained model with a different architecture and a knowledge cutoff of October 2023, to check that the finding is not model-specific.

We run the identical recall query and forecast prompt on GPT-4o-mini for all headline and earnings-call observations. The GPT-4o-mini $\text{LAP}^{P(\text{known})}$ distribution exhibits the same qualitative pattern as Llama: materially positive in-sample (mean 0.169 for headlines, 0.082 for earnings) and essentially zero post-cutoff (Figure VI), confirming that the cutoff collapse is a cross-model regularity (Tables A.8 and A.9). The cross-model LAP correlation is 0.50 for headlines and 0.42 for earnings, indicating that the two models memorize overlapping but not identical sets of firm-date outcomes.

Table VII reports the headline detection regression using GPT-4o-mini's own forecast signal and LAP on the in-sample window through October 2023. The interaction is positive and significant: $\hat{\delta} = 0.318$ ($t = 3.18$). Table VIII reports the earnings detection regression, where the interaction is $\hat{\delta} = 0.618$ ($t = 4.52$). The contamination finding replicates across model families in both applications.

4 Conclusion

Large language models are increasingly used as measurement instruments in empirical finance, extracting signals from text to predict stock returns, corporate investment, and macroeconomic outcomes. This paper shows that a substantial share of the apparent predictive content of these LLM-based measures reflects memorization of training-time material rather than genuine economic reasoning.

The contamination is task-specific and varies widely across domains. In the headline application of Lopez-Lira and Tang (2026), memorization amplifies the LLM forecast's marginal predictive power by about 32%; on the subset of firm-dates where the model shows no detectable directional recall, the headline signal loses all predictive power. In the earnings-call application of Jha, Qian, Weber, and Yang (2024), the amplification is about 13%, but the signal retains 94% of its predictive content on low-memorization firm-quarters, consistent with the earnings transcript carrying genuine economic information, though the paper cannot fully distinguish genuine reasoning from memorization through channels LAP does not reach. Two boundary cases complete the picture: U.S. CPI inflation is fully memorized, rendering the test mechanically powerless, while Chinese stocks (CSI 300) show no detectable directional recall, consistent with the model's English-centric training corpus.

The instrument that delivers these findings is a simple recall-based diagnostic built on a date-only query and a single interaction regression. The $\text{LAP}^{P(\text{known})}$ test requires only the ability to query the model and read off its first-token label probabilities, with no retraining and no access to proprietary training data. The contamination landscape we document is a fact about the current generation of widely used models and is important for the growing literature that treats LLM-extracted signals as valid economic measures. As LLMs become standard measurement instruments in economics and finance, the question is not whether they can extract predictive signals from text, but whether those signals reflect genuine economic content or memorized training-time outcomes. The diagnostic provides a low-cost audit for researchers who wish to assess, application by application, how much of their measurement instrument's apparent predictive content survives once memorization is netted out.

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Figure I. Example Prompts and Responses for Stock News Analysis

Panel A: Headline-direction prompt

Example Prompt:

Here is a piece of news: *"(2020-07-28) Kodak Triples on Loan to Make Covid-19 Drug Ingredients."*

Do you think this news is good, bad, or neutral for the stock price of *Eastman Kodak Company (KODK)* **in the short term?**

Write your answer as:
{**good / bad / neutral**}
{**confidence (0–1):**}
{**explanation (less than 25 words)**}

Example Response:

{**good**}
{**1.0**}
{**Loan to produce Covid-19 drug ingredients boosts prospects.**}

Panel B: Date-only recall query

Example Prompt:

On 2020-07-29, did the closing stock price of Eastman Kodak Company (KODK) go up or down compared to the previous trading day?

Answer based only on what you recall about *Eastman Kodak Company (KODK)* on that specific date. If you do not recall, answer "unknown".

Respond with exactly one word and nothing else: up, down, or unknown.

Example Response:

up

Figure II. Example Prompts and Responses for Earnings Call Analysis

Panel A: Capex-direction prompt

Example Prompt:

Here is an excerpt from the earnings call transcript of *Amazon.com Inc (AMZN)*:

Q1 2020 Amazon.com Inc Earnings Call (April 30, 2020)
"...We've invested more than $600 million in COVID-related costs in Q1 and expect these costs could grow to $4 billion or more in Q2. These include productivity headwinds in our facilities as we provide for social distancing and allow for the ramp-up of new employees, investments in personal protective equipment for employees, enhanced cleaning of our facilities, higher wages for our hourly teams, and hundreds of millions of dollars to develop COVID-19 testing capabilities...."

Do you think the company *Amazon.com Inc (AMZN)* **plans to increase or decrease its capital expenditures over the next year?**

Write your answer as:
{**significantly_increase / slightly_increase / no_change / slightly_decrease / significantly_decrease**}
{**confidence (0–1):**}
{**explanation (less than 25 words):**}

Example Response:

{**slightly_increase**}
{**0.8**}
{**Due to COVID-19 investments in safety measures and operations.**}

Panel B: Quarter-only recall query

Example Prompt:

In Q3 2020, did the capital expenditure of Amazon.com Inc (AMZN) increase or decrease compared to the previous quarter?

Answer based only on what you recall about *Amazon.com Inc (AMZN)* in that specific quarter. If you do not recall, answer "unknown".

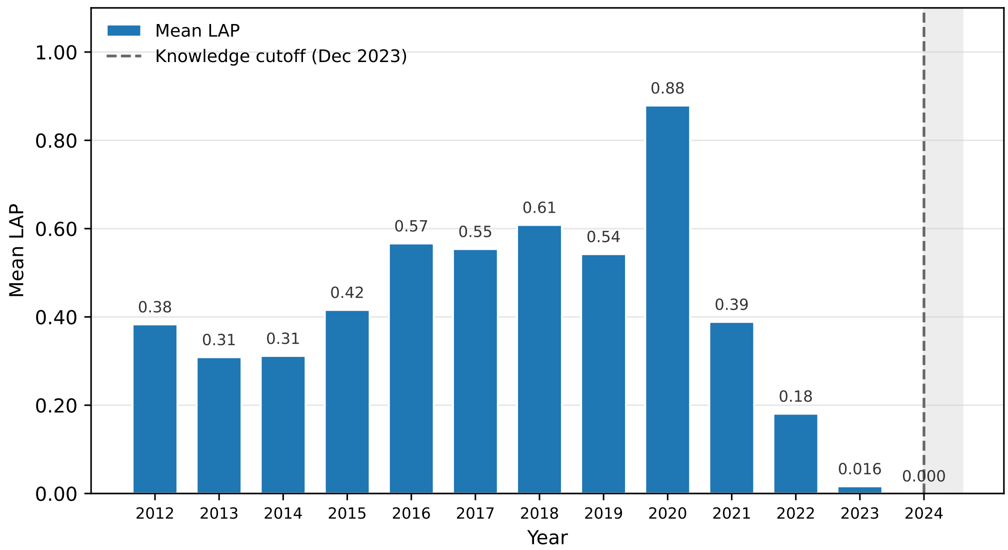
Respond with exactly one word: up, down, or unknown.

Example Response:

up

Figure III. Year-by-year evidence of $\text{LAP}^{P(\text{known})}$ collapse

Panel A: Stock News Prediction



Panel B: Earnings Call Prediction

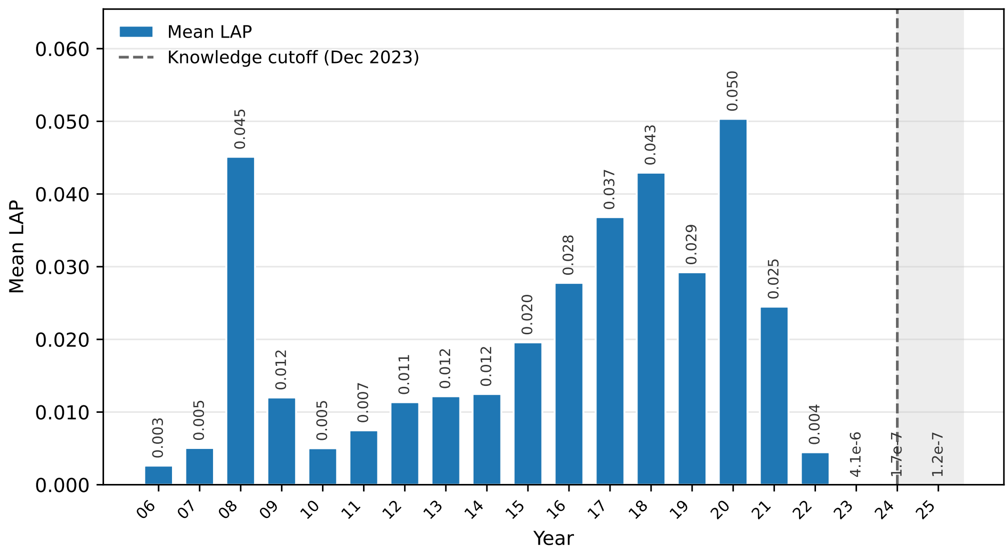


Figure IV. In-sample vs. out-of-sample distribution of $\text{LAP}^{P(\text{known})}$: Stock News Prediction

This figure shows the pooled within-window distribution of $\text{LAP}^{P(\text{known})}_{i,t+1}$ across firm-days, binned into five equal-width intervals on $[0, 1]$. Panel A pools the in-sample window (year 2012–2023); Panel B is the post-cutoff out-of-sample window (year 2024).

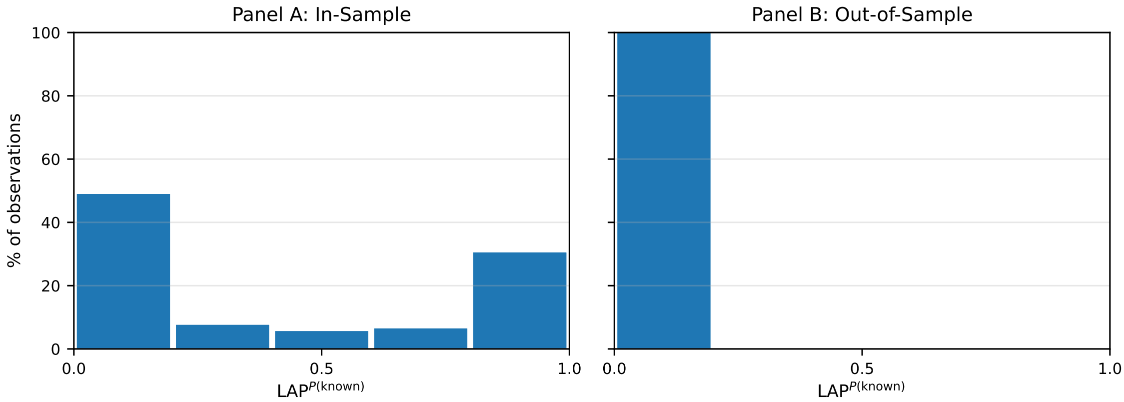


Figure V. In-sample vs. out-of-sample distribution of $\text{LAP}^{P(\text{known})}$: Earnings Call Prediction

This figure shows the pooled within-window distribution of $\text{LAP}^{P(\text{known})}_{i,q+2}$ across firm-quarters, binned into five equal-width intervals on $[0,1]$. The y-axis is capped at 5% so that the right-tail bins are visible; the leftmost bin $[0,0.2)$ saturates the cap. Panel A pools the in-sample window (2006Q1–2020Q4); Panel B is the post-cutoff out-of-sample window (2023Q3–2024Q1).

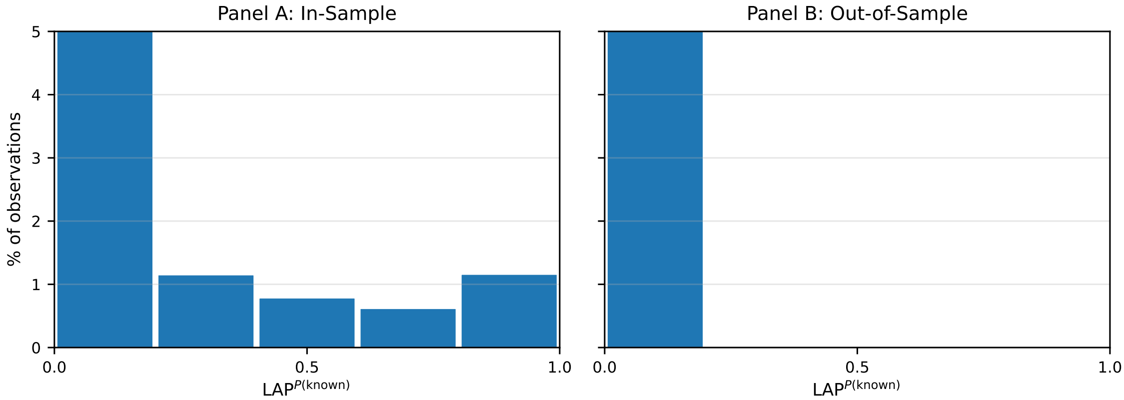
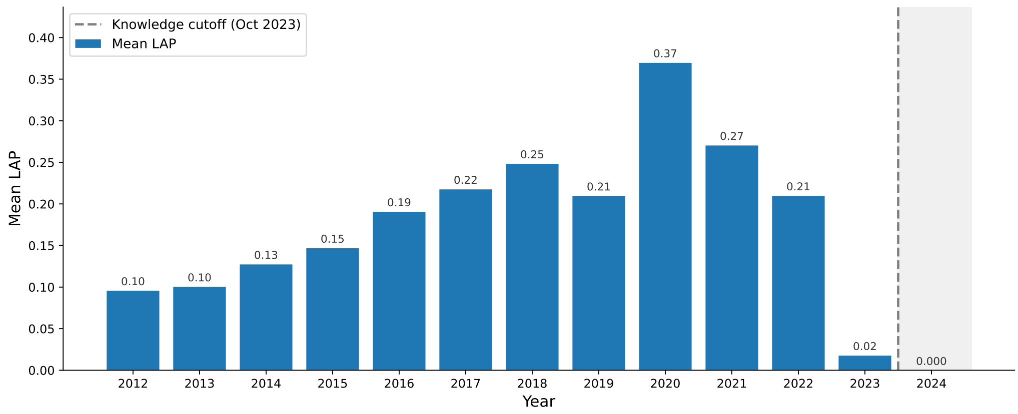


Figure VI. Year-by-year evidence of $\text{LAP}^{P(\text{known})}$ collapse: GPT-4o-mini

Panel A: Stock News Prediction



Panel B: Earnings Call Prediction

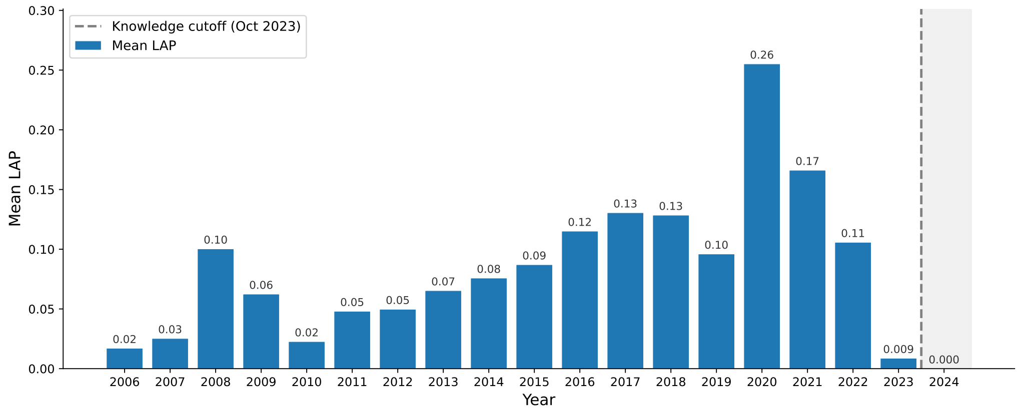


Table I. Regression of Next Day Returns on Answer Probabilities

This table reports regression results from the specification

$$r_{i,t+1} = \alpha_i + \lambda_t + \theta \cdot (U-D)_{i,t+1} + u_{i,t+1},$$

where $r_{i,t+1}$ is the next-day stock return for firm i at time t, measured in percentage points. The variable $(U-D)_{i,t+1}$ is the recall-query directional signal, equal to $P(\text{up}) - P(\text{down})$ from the Llama-3.3-70B date-only recall query at the next trading day. Column (1) reports the pooled estimate. Columns (2) and (3) split the in-sample window at the median of $\text{LAP}^{P(\text{known})}_{i,t+1}$, contrasting firm-days on which the model commits to a directional answer (high $\text{LAP}^{P(\text{known})}$) against firm-days on which it abstains (low $\text{LAP}^{P(\text{known})}$). All regressions include firm and date fixed effects, and standard errors are clustered by date. The sample includes all U.S. common stocks with at least one news headline covering the firm from January 2012 to December 2023.

	(1)	(2)	(3)
	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$
$(U-D)_{i,t+1}$	0.264***	0.301***	−0.190
	(3.53)	(3.55)	(−0.37)
Firm FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Mean $\text{LAP}^{P(\text{known})}$	0.412	0.785	0.040
N	91,361	45,639	45,229

Table II. Regression of Next Day Returns on LLM Prediction and LAP

This table reports regression results from the specification

$$r_{i,t+1} = \alpha_i + \lambda_t + \varphi\,\mathrm{LLM}_{i,t} + \eta\,\mathrm{LAP}^{P(\text{known})}_{i,t+1} + \delta\big(\mathrm{LLM}_{i,t} \times \mathrm{LAP}^{P(\text{known})}_{i,t+1}\big) + u_{i,t+1},$$

where $r_{i,t+1}$ is the next-trading-day return for firm i in percentage points, $\mathrm{LLM}_{i,t} \in \{-1, 0, +1\}$ is the Llama-3.3-70B headline-direction signal, and $\mathrm{LAP}^{P(\text{known})}_{i,t+1} = P(\text{up}) + P(\text{down})$ is the lookahead propensity from the date-only recall query at the next trading day. Column (2) adds the LAP main effect and its interaction with LLM. All specifications include firm and date fixed effects; standard errors are clustered by date. t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level. Sample starts from January 2012 to December 2023.

	(1)	(2)
	$r_{i,t+1}$	$r_{i,t+1}$
$\mathrm{LLM}_{i,t}$	0.209***	0.141***
	(12.18)	(7.55)
$\mathrm{LAP}^{P(\text{known})}_{i,t+1}$		0.024
		(0.22)
$\mathrm{LLM}_{i,t} \times \mathrm{LAP}^{P(\text{known})}_{i,t+1}$		0.162***
		(3.64)
Firm FE	Yes	Yes
Date FE	Yes	Yes
R^2	0.179	0.180
N	91,361	91,361

Table III. Regression of Next Day Returns on LLM Prediction, LAP, and Inner Confidence

This table reports regression results from the specification

$$r_{i,t+1} = \alpha_i + \lambda_t + \varphi \cdot \mathrm{LLM}_{i,t} + \eta \cdot \mathrm{LAP}^{P(\mathrm{known})}_{i,t+1} + \delta \cdot (\mathrm{LLM}_{i,t} \times \mathrm{LAP}^{P(\mathrm{known})}_{i,t+1}) + \eta' \cdot \mathrm{IC}_{i,t} + \delta' \cdot (\mathrm{LLM}_{i,t} \times \mathrm{IC}_{i,t}) + u_{i,t+1},$$

where $r_{i,t+1}$ is the next-trading-day return for firm i in percentage points, $\mathrm{LLM}_{i,t} \in \{-1, 0, +1\}$ is the Llama-3.3-70B headline-direction signal, $\mathrm{LAP}^{P(\mathrm{known})}_{i,t+1} = P(\mathrm{up}) + P(\mathrm{down})$ is the lookahead propensity from the date-only recall query at the next trading day, and $\mathrm{IC}_{i,t}$ (inner_confidence) is the joint label probability of the chosen direction in the headline-direction prompt and reflects the model's inner confidence (Chen, Didisheim, and Somoza, 2024). Column (1) replaces LAP with inner confidence and its interaction with the LLM signal. Column (2) horse-races the LAP and inner-confidence interactions jointly. All regressions include firm and date fixed effects, and standard errors are clustered by date. The sample includes all U.S. common stocks with at least one news headline covering the firm from January 2012 to December 2023.

	(1)	(2)
	$r_{i,t+1}$	$r_{i,t+1}$
$\mathrm{LLM}_{i,t}$	-0.387*	-0.459**
	(-1.81)	(-2.15)
$\mathrm{LAP}^{P(\mathrm{known})}_{i,t+1}$		0.023
		(0.21)
$\mathrm{LLM}_{i,t} \times \mathrm{LAP}^{P(\mathrm{known})}_{i,t+1}$		0.163***
		(3.64)
$\mathrm{IC}_{i,t}$	0.153	0.152
	(1.07)	(1.06)
$\mathrm{LLM}_{i,t} \times \mathrm{IC}_{i,t}$	0.602***	0.605***
	(2.79)	(2.81)
Firm FE	Yes	Yes
Date FE	Yes	Yes
R^2	0.179	0.180
N	91,357	91,357

Table IV. Regression of Future Capital Expenditure on Answer Probabilities

This table reports regression results from the specification

$$\text{CapEx}_{i,q+2} = \alpha_i + \lambda_q + \theta \cdot (U-D)_{i,q+2} + u_{i,q+2},$$

where $\text{CapEx}_{i,q+2}$ is the firm's capital-expenditure-to-assets ratio realized two quarters ahead, in percent, and $(U-D)_{i,q+2}$ is the recall-query directional signal, equal to $P(\text{up}) - P(\text{down})$ from the Llama-3.3-70B date-only capex-direction recall query at quarter $q+2$. Column (1) reports the pooled estimate. Columns (2) and (3) split firms at the median of their firm-level time-series average of $\text{LAP}^{P(\text{known})}_{i,q+2}$. All regressions include firm and quarter fixed effects, and standard errors are clustered by firm. The sample includes all firm-quarter observations from 2006Q1 to 2020Q4.

	(1)	(2)	(3)
	$\text{CapEx}_{i,q+2}$	$\text{CapEx}_{i,q+2}$	$\text{CapEx}_{i,q+2}$
$(U-D)_{i,q+2}$	0.310*	0.454**	103.424
	(1.84)	(2.57)	(1.70)
Firm FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Mean firm $\text{LAP}^{P(\text{known})}$	0.027	0.045	0.0001
N	116,518	69,461	47,057

Table V. Regression of Future Capital Expenditure on LLM Prediction and LAP

This table reports regression results from the specification

$$\text{CapEx}_{i,q+2} = \alpha_i + \lambda_q + \varphi \cdot \text{LLM}_{i,q} + \eta \cdot \text{LAP}^{P(\text{known})}_{i,q+2} + \delta \cdot \left(\text{LLM}_{i,q} \times \text{LAP}^{P(\text{known})}_{i,q+2}\right) + u_{i,q+2},$$

where $\text{CapEx}_{i,q+2}$ is the firm's capital-expenditure-to-assets ratio realized two quarters ahead, in percent, $\text{LLM}_{i,q} \in \{-1, -0.5, 0, +0.5, +1\}$ is the Llama-3.3-70B capex-direction signal extracted from the truncated earnings-call transcript at quarter q, and $\text{LAP}^{P(\text{known})}_{i,q+2} = P(\text{up}) + P(\text{down})$ is the lookahead propensity from the date-only capex-direction recall query at quarter $q+2$. Column (1) regresses $\text{CapEx}_{i,q+2}$ on the LLM signal alone. Column (2) adds the LAP main effect and its interaction with the LLM signal. All regressions include firm and quarter fixed effects, and standard errors are clustered by firm. The sample includes all firm-quarter observations from 2006Q1 to 2020Q4.

	(1)	(2)
	$\text{CapEx}_{i,q+2}$	$\text{CapEx}_{i,q+2}$
$\text{LLM}_{i,q}$	0.543***	0.529***
	(17.37)	(17.00)
$\text{LAP}^{P(\text{known})}_{i,q+2}$		0.052
		(0.28)
$\text{LLM}_{i,q} \times \text{LAP}^{P(\text{known})}_{i,q+2}$		0.580**
		(2.21)
Firm FE	Yes	Yes
Quarter FE	Yes	Yes
R^2	0.624	0.624
N	116,518	116,518

Table VI. Regression of Future Capital Expenditure on LLM Prediction, LAP, and Inner Confidence

This table reports regression results from the specification

$$\mathrm{CapEx}_{i,q+2} = \alpha_i + \lambda_q + \varphi \cdot \mathrm{LLM}_{i,q} + \eta \cdot \mathrm{LAP}^{P(\mathrm{known})}_{i,q+2} + \delta \cdot (\mathrm{LLM}_{i,q} \times \mathrm{LAP}^{P(\mathrm{known})}_{i,q+2}) + \eta' \cdot \mathrm{IC}_{i,q} + \delta' \cdot (\mathrm{LLM}_{i,q} \times \mathrm{IC}_{i,q}) + u_{i,q+2},$$

where $\mathrm{CapEx}_{i,q+2}$ is the firm's capital-expenditure-to-assets ratio realized two quarters ahead, in percent, $\mathrm{LLM}_{i,q} \in \{-1, -0.5, 0, +0.5, +1\}$ is the Llama-3.3-70B capex-direction signal extracted from the truncated earnings-call transcript at quarter q, $\mathrm{LAP}^{P(\mathrm{known})}_{i,q+2} = P(\mathrm{up}) + P(\mathrm{down})$ is the lookahead propensity from the date-only capex-direction recall query at quarter $q+2$, and $\mathrm{IC}_{i,q}$ (inner_confidence) is the joint label probability of the chosen direction in the capex-direction prompt and reflects the model's inner confidence (Chen, Didisheim, and Somoza, 2024). Column (1) replaces LAP with inner confidence and its interaction with the LLM signal. Column (2) horse-races the LAP and inner-confidence interactions jointly. All regressions include firm and quarter fixed effects, and standard errors are clustered by firm. The sample includes all firm-quarter observations from 2006Q1 to 2020Q4.

	(1)	(2)
	$\mathrm{CapEx}_{i,q+2}$	$\mathrm{CapEx}_{i,q+2}$
$\mathrm{LLM}_{i,q}$	0.218*	0.208*
	(1.81)	(1.72)
$\mathrm{LAP}^{P(\mathrm{known})}_{i,q+2}$		0.056
		(0.30)
$\mathrm{LLM}_{i,q} \times \mathrm{LAP}^{P(\mathrm{known})}_{i,q+2}$		0.574**
		(2.19)
$\mathrm{IC}_{i,q}$	−0.060	−0.058
	(−1.07)	(−1.03)
$\mathrm{LLM}_{i,q} \times \mathrm{IC}_{i,q}$	0.368***	0.363***
	(2.80)	(2.77)
Firm FE	Yes	Yes
Quarter FE	Yes	Yes
R^2	0.624	0.624
N	116,518	116,518

Table VII. Regression of Next-Day Return on GPT-4o-mini Prediction and LAP

This table reports the headline detection regression using GPT-4o-mini for both the forecast signal and the recall query. Column (1) regresses next-day returns on the GPT-4o-mini headline-direction signal alone. Column (2) adds the GPT-4o-mini $\text{LAP}^{P(\text{known})}$ main effect and its interaction with the LLM signal. The in-sample window is January 2012 to October 2023 (GPT-4o-mini's knowledge cutoff). Firm and date fixed effects are included; standard errors are clustered by date. t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level.

	(1)	(2)
	$r_{i,t+1}$	$r_{i,t+1}$
$\text{LLM}_{i,t}$	0.206***	0.157***
	(10.50)	(9.15)
$\text{LAP}^{P(\text{known})}_{i,t+1}$		0.063
		(0.48)
$\text{LLM}_{i,t} \times \text{LAP}^{P(\text{known})}_{i,t+1}$		0.318***
		(3.18)
Firm FE	Yes	Yes
Date FE	Yes	Yes
R^2	0.184	0.185
N	90,253	90,253

Table VIII. Regression of Future Capital Expenditure on GPT-4o-mini Prediction and LAP

This table reports the earnings detection regression using GPT-4o-mini for both the forecast signal and the recall query. Column (1) regresses two-quarters-ahead capital expenditure on the GPT-4o-mini capex-direction signal alone. Column (2) adds the GPT-4o-mini LAP$^{P(\text{known})}$ main effect and its interaction with the LLM signal. The in-sample window is 2006Q1 to 2020Q4; all recall quarters fall before GPT-4o-mini's October 2023 knowledge cutoff. Firm and quarter fixed effects are included; standard errors are clustered by firm. t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level.

	(1)	(2)
	$\text{CapEx}_{i,q+2}$	$\text{CapEx}_{i,q+2}$
$\text{LLM}_{i,q}$	0.654***	0.563***
	(15.64)	(13.58)
$\text{LAP}^{P(\text{known})}_{i,q+2}$		−0.328***
		(−3.13)
$\text{LLM}_{i,q} \times \text{LAP}^{P(\text{known})}_{i,q+2}$		0.618***
		(4.52)
Firm FE	Yes	Yes
Quarter FE	Yes	Yes
R^2	0.624	0.624
N	116,531	116,531

Online Appendix

A Procedure for Applying the Lookahead-Bias Test

This appendix gives the test as a sequence of actions: each step ends either in a conclusion, in which case the researcher stops, or in a handoff to the next step. The setting is a generic forecasting application in which a researcher observes text data $X_{i,t}$ for entities i (firms, industries, countries, or assets) and uses an LLM to predict an outcome $Y_{i,t+h}$ at horizon $h \geq 1$. Three ingredients are required: a panel of text observations with matched realized outcomes; the training-data cutoff date of the LLM, typically disclosed in the model card; and access to token-level probabilities at the answer position, which open-weight models provide directly and some commercial APIs expose as log-probabilities (when unavailable, answer frequencies across repeated samples at positive temperature are a coarser substitute). The procedure is also specified in a `CLAUDE.md` instruction file that runs it end-to-end with Claude Code, available for download here.

Step 1: Generate the forecast and test post-cutoff performance first. Query the LLM with the text $X_{i,t}$ and ask for a directional prediction of $Y_{i,t+h}$ from a fixed label set, with temperature zero; map the label to a numeric signal $\hat{\mu}_{i,t}$. Generate the forecast exactly as in the intended application, for observations both before and after the model's training cutoff: the object under test is the researcher's own pipeline. Then estimate predictive performance on the post-cutoff sample alone.

- **If post-cutoff performance is significant: stop.** The application demonstrably predicts on data the model had no training-set exposure to, so genuine skill is established for that model version and window. This presumes the model is frozen at its stated cutoff, with no later fine-tuning or retrieval, and that the window is economically representative. Our headline application is of this type. If full-sample magnitudes are also reported, Steps 2 to 4 quantify how much of them reflects memorization.

- **Otherwise, continue.** Post-cutoff windows are often short, so an insignificant post-cutoff result alongside significant in-sample performance is common and does not by itself separate skill from memorization.

Step 2: Measure memorization and compare it across the cutoff. Construct the date-only recall query for the realization period: supply only the entity identifier and the target date or quarter of $Y_{i,t+h}$, with no text, no fundamentals, and no other contemporaneous context, and ask whether the outcome went up or down, with an explicit `unknown` option (Section 2.1). From the token probabilities at the answer position, compute

$$\text{LAP}_{i,t+h}^{P(\text{known})} = P(\text{up}) + P(\text{down}), \qquad (U-D)_{i,t+h} = P(\text{up}) - P(\text{down}).$$

The recall date must match the realization horizon of the outcome, not the date of the text, and each answer label should be a single token under the model's tokenizer. Then compare the LAP distribution in and out of sample, for instance at the level of yearly means.

- **If LAP is approximately zero on both sides of the cutoff: stop.** Two readings are observationally equivalent: the model has not memorized the outcomes, or the date-only recall query cannot elicit the memorization that exists in this domain and model, whether because of the query's form, refusal behavior, or safety training. We flag this explicitly as a limitation of the method: a silent recall query renders the detection step uninformative rather than exonerating, and certification rests on post-cutoff evidence as it accumulates.

- **If in-sample LAP is bounded away from zero and collapses toward zero post-cutoff: continue.** Under the frozen-model assumption this pattern indicates that the recall query captures memorization rather than salience or guessing: it goes silent exactly where memory is impossible.

Step 3: Check that the memorized content is outcome-relevant. Regress the realized outcome on the recall signal with entity and realization-period fixed effects,

$$Y_{i,t+h} = \alpha_i + \lambda_{t+h} + \theta \cdot (U-D)_{i,t+h} + u_{i,t+h},$$

on the pooled pre-cutoff sample and on subsamples split at the median of $\text{LAP}^{P(\text{known})}$ or of its entity-level average, following Equations (8) and (10). Because the recall query is outcome-blind by construction, a positive θ on the high-LAP subsample together with a null on the low-LAP subsample establishes that the model has memorized outcome-relevant content. Continue.

Step 4: Test whether memorization leaks into the forecast. Estimate Equation (7),

$$Y_{i,t+h} = \beta_1 \hat{\mu}_{i,t} + \beta_2 \, \mathrm{LAP}_{i,t+h} + \beta_3 \, (\mathrm{LAP}_{i,t+h} \times \hat{\mu}_{i,t}) + \epsilon_{i,t+h},$$

with the controls, fixed effects, and clustering appropriate to the panel. The one-sided test of lookahead bias is $\beta_3 > 0$: the forecast's predictive content is amplified precisely where memorization is strong (Proposition 1). As a placebo, β_3 should be statistically indistinguishable from zero post-cutoff.

- **If β_3 is insignificant: stop.** Memorization exists (Steps 2 and 3), yet we detect no reliance of the forecast on it. The conclusion is that the predictions show no detectable reliance on directional recall memorization, which is the channel the recall query elicits and the interaction tests. A small undetected effect cannot be excluded, and neither can other forms of memorization. Three specific sources of false negatives deserve emphasis. First, *cue-dependent leakage*: the full headline or transcript may activate memorized post-event coverage through contextual cues (event descriptions, industry terms, temporal references) that the stripped date-only recall query does not trigger, so the forecast can be contaminated even where LAP ≈ 0. Second, *correlated-quantity leakage*: the recall query probes one specific variable at one horizon, and memorization of correlated quantities (sector returns, earnings surprises, macro announcements) at other horizons could contaminate the forecast without being detected by the matched recall query. Third, *homogeneous leakage*: when memorization is approximately uniform within a period, the contamination is absorbed by time fixed effects and the LLM main effect, leaving no residual variation for the interaction to exploit (the CPI application illustrates this limiting case). All three channels are bounded only by the post-cutoff comparison.

- **If β_3 is significantly positive: the application is contaminated.** The forecast's predictive content loads in part on memorized outcomes. The researcher should report this finding and, where feasible, restrict evaluation to the post-cutoff window.

Recommended reporting checklist. For any paper that uses a modern LLM to interpret historical financial text, the following disclosures help readers assess the point-in-time validity of the results:

1. *Model card*: model version, knowledge cutoff date, inference settings (temperature, top-k, batch size), and the exact prompt template.

2. *Baseline performance*: the raw LLM predictive power on the full sample and on the post-cutoff window.

3. *Recall audit*: the $\text{LAP}^{P(\text{known})}$ distribution (in-sample vs. post-cutoff), the $(U-D)$ validation regression, and the year-by-year cutoff collapse.

4. *Detection result*: the interaction coefficient $\hat{\beta}_3$ with the minimum detectable effect δ_{MDE} and maximum contamination share MCS_{MDE} to calibrate power.

5. *Conservative performance*: restrict the sample to observations with LAP below the maximum post-cutoff value and re-estimate the LLM-only specification. This low-LAP subsample estimate provides a conservative benchmark for forecasting performance in the absence of memorization.

B Proofs of Theoretical Results

B.1 Proof of Proposition 1

Proposition 1 maintains the following benchmark conditions, where C_t denotes the controls and fixed effects included in Equation (7) and all moments below condition on C_t:

(A1) *LAP exogeneity to the signal.* LAP_t is independent of $\mu(X_t)$ given C_t.

(A2) *Homoskedasticity.* $\mathrm{Var}(\varepsilon_{t+1} \mid \mathrm{LAP}_t, C_t) = \sigma_\varepsilon^2$.

(A3) *LAP exogeneity to the innovation.* LAP_t is independent of ε_{t+1} given C_t.

(A4) *Partial transmission.* $0 \le \gamma < 1$, excluding perfect transmission.

(A5) *Innovation moments.* $\mathbb{E}[\varepsilon_{t+1} \mid \mu(X_t), \mathrm{LAP}_t, C_t] = 0$ and $\mathbb{E}[\varepsilon_{t+1}^2 \mid \mu(X_t), \mathrm{LAP}_t, C_t] = \sigma_\varepsilon^2$.

In (A3), the $t+1$ subscript on $\mathrm{LAP}_{i,t+1}^{P(\text{known})}$ indexes the *target date* of the recall query, not its information set: the query uses only the firm name, ticker, and target date, with no real-time access to ε_{t+1}. Empirically, (A3) is supported by the post-cutoff placebo, where LAP_t collapses to essentially zero and β_3 is statistically indistinguishable from zero.

Proof. Throughout, let $\mathrm{LAP} \equiv \mathrm{LAP}_{i,t+1}^{P(\text{known})}$ and adopt the cardinal measurement baseline $L_t = \mathrm{LAP}_t$, so that $L_t \in [0,1]$ is observable. The DGP is $Y_{t+1} = \mu(X_t) + \varepsilon_{t+1}$ with $\mu(X_t) \perp \varepsilon_{t+1}$, and the contaminated forecast is $\hat{\mu}_t = \mu(X_t) + \gamma\,\mathrm{LAP}_t\,\varepsilon_{t+1}$, where $\gamma \in [0,1)$ is the scalar contamination loading. All moments throughout the proof condition on C_t, which we suppress for notational simplicity.

Part (a): Under $\gamma = 0$. With $\gamma = 0$, $\hat{\mu}_t = \mu(X_t)$ and $Y_{t+1} - \hat{\mu}_t = \varepsilon_{t+1}$. By (A5), $\mathbb{E}[\varepsilon_{t+1} \mid \mu(X_t), \mathrm{LAP}_t] = 0$; since $\hat{\mu}_t = \mu(X_t)$, this gives $\mathbb{E}[\varepsilon_{t+1} \mid \hat{\mu}_t, \mathrm{LAP}_t] = 0$. The interaction $\hat{\mu}_t \cdot \mathrm{LAP}_t$ is a measurable function of $(\hat{\mu}_t, \mathrm{LAP}_t)$, so $\mathbb{E}[\varepsilon_{t+1} \mid \hat{\mu}_t \cdot \mathrm{LAP}_t, \hat{\mu}_t, \mathrm{LAP}_t] = 0$. Standard OLS arguments give $\beta_3 = 0$.

Part (b): Under $\gamma \in (0,1)$. Substituting $\hat{\mu}_t = \mu(X_t) + \gamma\,\mathrm{LAP}_t\,\varepsilon_{t+1}$ into the OLS slope of Y_{t+1} on $\hat{\mu}_t$ within a fixed value of $\mathrm{LAP}_t = a$:

$$\begin{aligned} \beta(a) &\equiv \frac{\mathrm{Cov}(Y_{t+1}, \hat{\mu}_t \mid \mathrm{LAP}_t = a)}{\mathrm{Var}(\hat{\mu}_t \mid \mathrm{LAP}_t = a)} \\ &= \frac{\mathrm{Var}(\mu(X_t) \mid \mathrm{LAP}_t = a) + \gamma\, a\, \mathrm{Var}(\varepsilon_{t+1} \mid \mathrm{LAP}_t = a)}{\mathrm{Var}(\mu(X_t) \mid \mathrm{LAP}_t = a) + \gamma^2 a^2\, \mathrm{Var}(\varepsilon_{t+1} \mid \mathrm{LAP}_t = a)}, \end{aligned} \tag{12}$$

where the conditional cross-term $\mathrm{Cov}(\mu(X_t), \varepsilon_{t+1} \mid \mathrm{LAP}_t) = 0$ follows from (A5) by iterated expectations ($\mathbb{E}[\varepsilon_{t+1} \mid \mu(X_t), \mathrm{LAP}_t] = 0$ implies $\mathbb{E}[\mu(X_t)\,\varepsilon_{t+1} \mid \mathrm{LAP}_t] = 0$), and (A5) further provides $\mathbb{E}[\varepsilon_{t+1}^2 \mid \mu(X_t), \mathrm{LAP}_t] = \sigma_\varepsilon^2$ for the second-moment terms.

Under (A1) and (A2), $\mathrm{Var}(\mu(X_t) \mid \mathrm{LAP}_t = a) = \sigma_\mu^2$ and $\mathrm{Var}(\varepsilon_{t+1} \mid \mathrm{LAP}_t = a) = \sigma_\varepsilon^2$ do not depend on a, so Equation (12) reduces to

$$\beta(a) = \frac{\sigma_\mu^2 + \gamma\, a\, \sigma_\varepsilon^2}{\sigma_\mu^2 + \gamma^2\, a^2\, \sigma_\varepsilon^2}.$$

Two properties of $\beta(a)$ are immediate. First, $\beta(0) = 1$. Second, for $a \in (0,1]$ and $\gamma \in (0,1)$, $\beta(a) > 1$: since $\gamma < 1$ implies $\gamma > \gamma^2$, the numerator $\sigma_\mu^2 + \gamma a \sigma_\varepsilon^2$ exceeds the denominator $\sigma_\mu^2 + \gamma^2 a^2 \sigma_\varepsilon^2$ whenever $a \in (0,1]$. Differentiating, $\partial\beta(a)/\partial a|_{a=0} = \gamma\,\sigma_\varepsilon^2/\sigma_\mu^2 > 0$, so $\beta(a)$ is strictly increasing in LAP in a neighborhood of zero.

The detection regression in Equation (7) approximates the conditional slope $\beta(a)$ with the linear form $\beta_1 + \beta_3 \cdot a$, so β_3 corresponds to the OLS-implicit best-linear-fit slope of $\beta(a)$ over the support of LAP_t. By standard OLS properties, β_3 is a *variance-weighted* covariance between $\beta(\mathrm{LAP})$ and LAP with positive weights $w(a) \propto \mathrm{Var}(\hat{\mu}_t \mid \mathrm{LAP}_t = a)$:

$$\beta_3 \;=\; \frac{\mathrm{Cov}_w\big(\beta(\mathrm{LAP}), \mathrm{LAP}\big)}{\mathrm{Var}_w(\mathrm{LAP})}.$$

Since the weights $w(a)$ are strictly positive and $\beta(a) > \beta(0) = 1$ for every $a > 0$ under (A4), $\beta_3 > 0$ holds whenever $\beta(a)$ is non-decreasing and non-constant on the support of LAP_t. Proposition 1 states this monotonicity as a maintained, empirically checkable condition; it can be assessed by estimating the OLS slope of Y_{t+1} on $\hat{\mu}_t$ within LAP bins.

Boundary case $\gamma = 1$. If $\gamma = 1$, then $\hat{\mu}_t = \mu(X_t) + \mathrm{LAP}_t\, \varepsilon_{t+1}$ and the conditional slope reduces to $\beta(a) = (\sigma_\mu^2 + a\,\sigma_\varepsilon^2)/(\sigma_\mu^2 + a^2\,\sigma_\varepsilon^2)$. This equals 1 at the endpoints $a = 0$ (where $\hat{\mu}_t = \mu(X_t)$) and $a = 1$ (where $\hat{\mu}_t = Y_{t+1}$), but exceeds 1 in the interior $a \in (0,1)$. Whether β_3 remains informative therefore depends on the distribution of LAP_t: it captures the interior amplification when LAP_t has continuous mass, but vanishes mechanically when LAP_t is concentrated at the endpoints $\{0,1\}$. (A4) excludes this boundary; see Section 2.2. □

B.2 When Does the Monotonicity Condition Hold? A Regime-Complete Characterization

Proposition 1(b) maintains that the conditional slope $\beta(a)$ is non-decreasing and non-constant on the support of LAP_t. Table A.10 validates this condition empirically: for earnings calls, the within-tercile slope is strictly increasing (0.402 [0.316, 0.488], 0.478 [0.394, 0.562], 0.754 [0.638, 0.870]); for headlines, the bottom two terciles have overlapping 95% confidence intervals (0.171 [0.124, 0.218] and 0.151 [0.110, 0.192]), while the top tercile is clearly separated (0.302 [0.229, 0.375]). The profile is clearly non-constant in both applications, and by part (b) below, non-constancy is inconsistent with $\gamma = 0$. This subsection makes that condition primitive: it characterizes exactly when it holds as a function of the contamination loading γ, the noise-to-signal ratio, and the LAP support; establishes that the null $\gamma = 0$ is equivalent to a *constant* conditional-slope profile; and derives how to read the detection statistic when the monotone condition fails. Throughout, the benchmark conditions (A1)–(A3) and (A5) of Appendix B.1 hold, all moments condition on C_t, and $R \equiv \sigma_\varepsilon^2/\sigma_\mu^2$ denotes the noise-to-signal ratio.

Proposition 2 (Regime-Complete Characterization)**.** *Under the contamination model (5) and the benchmark conditions of Appendix B.1:*

(a) (Shape of the profile.) *For $\gamma > 0$, the conditional slope in Equation (12) satisfies*

$$\operatorname{sign}\beta'(a) \;=\; \operatorname{sign}\Big[\,\sigma_\mu^2(1-2\gamma a) \,-\, \gamma^2 a^2 \sigma_\varepsilon^2\,\Big],$$

so $\beta(a)$ is strictly increasing on $[0, a^*)$ and strictly decreasing beyond a^*, where

$$a^* \;=\; \frac{\sqrt{1+R}-1}{\gamma R}.$$

The monotonicity condition maintained in Proposition 1(b) therefore holds on a support $[0, \bar{a}]$ if and only if $a^* \geq \bar{a}$; on the full support $[0, 1]$, if and only if

$$\gamma \;\leq\; \bar{\gamma} \;\equiv\; \frac{\sqrt{1+R}-1}{R} \;\leq\; \tfrac{1}{2}.$$

(b) (Sharp null.) *$\gamma = 0$ if and only if $\beta(a)$ is constant in a (in which case $\beta \equiv 1$). Consequently, under the maintained exogeneity conditions, a statistically significant $\hat{\beta}_3$ of* either *sign is inconsistent with the joint null* $\{\gamma = 0$ *and (A1), (A3), (A5)*$\}$*. A negative* $\hat{\beta}_3$ *is not itself read as detection: within the model it can arise only in the regime of part (c), and outside the model it can reflect a failure of the exogeneity conditions; in either case it calls for the profile diagnostics below.*

(c) (Beyond the peak.) *If* $\gamma > \bar{\gamma}$ *and* LAP_t *places positive mass above* a^**, the best-linear-fit coefficient* β_3*, the variance-weighted covariance of Appendix B.1, can be biased toward zero, or negative, even though* $\gamma > 0$*. Because part (b) is an equivalence at the level of the profile* $\beta(\cdot)$*, the operative test of* $\gamma = 0$ *in this regime is* constancy of the profile*: estimate within-bin slopes* $\hat{\beta}(a_k)$ *by interacting* $\hat{\mu}_t$ *with LAP-bin indicators and test equality across bins (Wald test, clustered standard errors). The linear coefficient* β_3 *is the profile's weighted linear summary, and its one-sided interpretation in Proposition 1(b) applies exactly when the estimated profile is non-decreasing.*

Proof. *(a)* Differentiating $\beta(a) = (\sigma_\mu^2 + \gamma a \sigma_\varepsilon^2)/(\sigma_\mu^2 + \gamma^2 a^2 \sigma_\varepsilon^2)$, the numerator of $\beta'(a)$ is

$$\gamma\sigma_\varepsilon^2\left(\sigma_\mu^2 + \gamma^2 a^2 \sigma_\varepsilon^2\right) - \left(\sigma_\mu^2 + \gamma a \sigma_\varepsilon^2\right) 2\gamma^2 a \sigma_\varepsilon^2 = \gamma\sigma_\varepsilon^2\left[\sigma_\mu^2(1-2\gamma a) - \gamma^2 a^2 \sigma_\varepsilon^2\right],$$

which gives the sign statement. The bracketed quadratic $q(a) = \sigma_\mu^2(1-2\gamma a) - \gamma^2 a^2 \sigma_\varepsilon^2$ satisfies $q(0) = \sigma_\mu^2 > 0$ and has a unique positive root, obtained from $\gamma^2\sigma_\varepsilon^2 a^2 + 2\gamma\sigma_\mu^2 a - \sigma_\mu^2 = 0$ as $a^* = \left[\sqrt{\sigma_\mu^4 + \sigma_\mu^2\sigma_\varepsilon^2} - \sigma_\mu^2\right]/(\gamma\sigma_\varepsilon^2) = (\sqrt{1+R} - 1)/(\gamma R)$. Setting $a^* \geq 1$ and solving for γ gives $\bar{\gamma}$; the map $R \mapsto (\sqrt{1+R} - 1)/R$ is decreasing with limit $1/2$ as $R \downarrow 0$, whence $\bar{\gamma} \leq 1/2$.

(b) If $\gamma = 0$, Equation (12) gives $\beta(a) \equiv 1$. Conversely, if $\gamma > 0$, then $q(0) = \sigma_\mu^2 > 0$ and q is a nontrivial quadratic, so $\beta'(a) > 0$ on a neighborhood of zero and $\beta(\cdot)$ is non-constant. The inferential statement follows by combining this equivalence with Proposition 1(a), under which the population β_3 is zero whenever $\gamma = 0$ and the exogeneity conditions hold: a nonzero population β_3 of either sign therefore requires either $\gamma > 0$ or a violated benchmark condition.

(c) Immediate from the weighted-covariance representation: if $\beta(\cdot)$ increases below a^* and decreases above it, and LAP_t has sufficient mass beyond a^*, the weighted covariance $\text{Cov}_w(\beta(\text{LAP}), \text{LAP})$ can be zero or negative despite $\beta(\cdot)$ being non-constant. □

Remark 1 (Identification strength: when does LAP variation suffice?)**.** *The detection coefficient* β_3 *is identified from residual variation in the interaction* $\hat{\mu}_t \times LAP_t$ *after absorbing the main effects and fixed effects. By Frisch–Waugh–Lovell,* $\hat{\beta}_3 = Cov(\tilde{Y}, \tilde{W})/\text{Var}(\tilde{W})$*, where*

$\tilde{W}$ *is* $\hat{\mu}_t \times LAP_t$ *residualized on* $\{\hat{\mu}_t, LAP_t, \alpha_i, \lambda_t\}$*, and* $SE(\hat{\beta}_3) \approx \sigma_u/\sqrt{N_{\mathit{eff}}\mathrm{Var}(\tilde{W})}$ *under cluster-robust inference with* N_{eff} *effective observations. As LAP concentrates at a single value, either near zero (no memorization detected) or near one (saturation),* $\mathrm{Var}(\tilde{W}) \to 0$ *and the interaction becomes collinear with the main effect of* $\hat{\mu}_t$*. The test fails not because contamination is absent, but because the design lacks the variation to detect it.*

Two diagnostics characterize the design's power prior to estimation. First, the variance inflation factor for the interaction, $VIF_3 = 1/(1-R^2_W)$*, where* R^2_W *is from regressing* $\hat{\mu}_t \times LAP_t$ *on* $\{\hat{\mu}_t, LAP_t, FE\}$*, directly measures degeneracy. A bimodal LAP distribution (mass at 0 and mass at 1, as in our headline application) yields low* VIF_3 *and is favorable for identification; concentration at either extreme drives* VIF_3 *upward. Second, the minimum detectable effect at power* $1-\pi$ *and size* α *is*

$$\delta_{MDE} = (z_{1-\alpha} + z_{1-\pi}) \frac{\sigma_u}{\sqrt{N_{\mathit{eff}}\,\mathrm{Var}(\tilde{W})}},$$

which can be expressed in the paper's economic units as the minimum boost to forecast performance per one standard deviation of LAP: $\rho_{MDE} = \delta_{MDE} \cdot SD(LAP)/\hat{\phi}$*, where* $\hat{\phi}$ *is the estimated forecast slope from Equation (7). If* ρ_{MDE} *exceeds an economically meaningful threshold (e.g., 25% of the forecast's predictive content), the design is underpowered for the contamination question, and the researcher should report this alongside the null and refrain from interpreting* $\hat{\beta}_3 \approx 0$ *as evidence against contamination.*

Because LAP is itself a noisy proxy for latent memorization, there is an additional source of downward bias beyond low variation. Classical errors-in-variables bias pushes $\hat{\beta}_3$ *toward zero even when* $\mathrm{Var}(\tilde{W})$ *is adequate, and this bias cannot be corrected without a second, independent measurement of memorization. Accordingly, an insignificant* $\hat{\beta}_3$ *is a one-sided result: it is consistent with no contamination, but it is also consistent with contamination whose detection is weakened by measurement error.*

Remark 2 (Forecast noise widens the monotone regime but qualifies the null)**.** *The baseline model (5) abstracts from forecast noise. Writing* $\hat{\mu}_t = \mu(X_t) + \gamma L_t \varepsilon_{t+1} + \eta_t$ *with* η_t *mean-zero noise,* $\mathrm{Var}(\eta_t \mid LAP_t, C_t) = \sigma^2_\eta$*, uncorrelated with* $(\mu(X_t), \varepsilon_{t+1})$*, the conditional slope becomes*

$$\beta(a) = \frac{\sigma^2_\mu + \gamma\, a\, \sigma^2_\varepsilon}{\sigma^2_\mu + \sigma^2_\eta + \gamma^2 a^2 \sigma^2_\varepsilon}, \qquad a^* = \frac{\sqrt{\sigma^4_\mu + \sigma^2_\varepsilon(\sigma^2_\mu + \sigma^2_\eta)} - \sigma^2_\mu}{\gamma\, \sigma^2_\varepsilon},$$

with $\beta(0) = \sigma_\mu^2/(\sigma_\mu^2 + \sigma_\eta^2)$ and $\bar{\gamma} = \gamma\, a^*$ now increasing *in σ_η^2: noisier forecast signals sit deeper inside the monotone regime, which is the empirically relevant case for noisy directional signals such as ours. Two qualifications follow. First, with $\sigma_\eta^2 > 0$ the null profile is the constant $\sigma_\mu^2/(\sigma_\mu^2+\sigma_\eta^2)$, so Proposition 1(a) additionally requires that* $\mathrm{Var}(\mu(X_t) \mid LAP_t, C_t)$ *and* $\mathrm{Var}(\eta_t \mid LAP_t, C_t)$ *not vary with LAP_t, i.e., that the signal-to-noise ratio of the forecast not covary with memorization (for instance, through heavily covered firms being both better memorized and easier to forecast). Variation in* $\mathrm{Var}(\varepsilon_{t+1} \mid LAP_t, C_t)$ *remains harmless under the null, since ε_{t+1} does not enter $\hat{\mu}_t$ when $\gamma = 0$. Interacting $\hat{\mu}_t$ with volatility, size, and attention controls in Equation (7) diagnoses this condition. Second, these moment conditions are checkable out of sample: post-cutoff, $LAP_t \approx 0$ implies $\hat{\mu}_t = \mu(X_t) + \eta_t$, so under second-moment stationarity across the training cutoff the post-cutoff joint moments identify $\sigma_\mu^2 = Cov(Y_{t+1}, \hat{\mu}_t)$, $\sigma_\varepsilon^2 =$* $\mathrm{Var}(Y_{t+1}) - \sigma_\mu^2$*, and $\sigma_\eta^2 =$* $\mathrm{Var}(\hat{\mu}_t) - \sigma_\mu^2$*; the residual variance* $\mathrm{Var}(Y_{t+1} - \hat{\mu}_t) = \sigma_\varepsilon^2 + \sigma_\eta^2$ *alone does not separate the last two. These moments deliver a computable $\bar{\gamma}$ against which the estimated within-bin slope profile can be read.*

C Tables

Table A.1. Variable Definitions

This table summarizes the construction and definitions of all variables used in the analysis.

Variable	Definition
	News Headlines Predicting Stock Returns
$r_{i,t+1}$	Next-trading-day return for firm i following headline date t, in percentage points.
$\text{LLM}_{i,t}$	Headline-direction signal extracted from Llama-3.3-70B's response to the news headline observed at time t. Takes value $+1$ for an "up" prediction, -1 for "down", and 0 otherwise.
$\text{LAP}_{i,t+1}^{P(\text{known})}$	Lookahead propensity at the next trading day, computed as $P(\text{up})+P(\text{down})$ from Llama-3.3-70B's date-only recall query asking the model whether the firm's price went up or down on day $t+1$.
$(U-D)_{i,t+1}$	Recall-query directional signal, equal to $P(\text{up})-P(\text{down})$ from the same date-only recall query at the next trading day $t+1$.
$\text{inner\_confidence}_{i,t}$	Joint label probability assigned by Llama-3.3-70B to its chosen answer in the headline-direction prompt at time t, equal to $P(\text{good})$ when the model's response is `good`, $P(\text{neutral})$ when `neutral`, and $P(\text{bad})$ when `bad`. It reflects the model's inner confidence (Chen, Didisheim, and Somoza, 2024).

Continued on next page

Variable	Definition
	Earnings Call Predicting Capital Expenditure
$\text{CapEx}_{i,q+2}$	Firm i's capital-expenditure-to-assets ratio realized two quarters after the earnings-call quarter q, in percent. Winsorized at the 1% / 99% level within each year-quarter yq.
$\text{LLM}_{i,q}$	Capex-direction signal extracted from Llama-3.3-70B's response to the truncated earnings-call transcript at quarter q. Takes values in $\{-1, -0.5, 0, +0.5, +1\}$, where higher values indicate a more positive predicted change in capital expenditure.
$\text{LAP}^{P(\text{known})}_{i,q+2}$	Lookahead propensity at quarter $q+2$, computed as $P(\text{up}) + P(\text{down})$ from Llama-3.3-70B's date-only capex-direction recall query at quarter $q+2$.
$(U-D)_{i,q+2}$	Recall-query directional signal for capex direction at $q+2$, equal to $P(\text{up}) - P(\text{down})$ from the same recall query.
$\text{inner\_confidence}_{i,q}$	Joint label probability assigned by Llama-3.3-70B to its chosen answer in the capex-direction prompt at quarter q, equal to $P(\text{label})$ where label matches the model's response among {significantly decrease, slightly decrease, no change, slightly increase, significantly increase}. It reflects the model's inner confidence (Chen, Didisheim, and Somoza, 2024).

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Variable	Definition
	CPI Inflation Prediction
$\Delta\pi_m^{\text{YoY}}$	Month-over-month change in U.S. year-over-year CPI inflation at month m, in percentage points. Constructed from the seasonally adjusted CPI-U (CPIAUCSL) series from FRED.
LLM_m	Inflation-direction signal extracted from Llama-3.3-70B's response to a date-only prompt asking the model to forecast the direction of change in year-over-year CPI inflation at month m. Takes values in $\{-1, -0.5, 0, +0.5, +1\}$.
$\text{LAP}_{m+1}^{P(\text{known})}$	Lookahead propensity at month $m+1$, computed as $P(\text{up}) + P(\text{down})$ from Llama-3.3-70B's date-only recall query asking whether the year-over-year CPI inflation rate went up or down in month $m+1$ compared to the previous month.

Continued on next page

Variable	Definition
	Chinese Stock Market (CSI 300)
$r_{i,t+1}$	Next-trading-day return for CSI 300 constituent i following date t, in percentage points.
$\text{LAP}^{P(\text{known})}_{i,t+1}$	Lookahead propensity at the next trading day, computed as $P(\text{up}) + P(\text{down})$ from Llama-3.3-70B's date-only recall query asking the model whether the stock's closing price went up or down on day $t+1$.
$(U-D)_{i,t+1}$	Recall-query directional signal, equal to $P(\text{up})-P(\text{down})$ from the same date-only recall query at the next trading day $t+1$.

Table A.3. Summary Statistics: Stock News Prediction

Panel A: In-sample

This panel presents summary statistics for the variables used in the stock-news analysis on the in-sample window (January 2012 to December 2023). $\text{LAP}^{P(\text{known})}$ is the lookahead propensity constructed as $P(\text{up})+P(\text{down})$ from the Llama-3.3-70B date-only recall query, prompted on the next trading day after each headline. $r_{i,t+1}$ is the next-trading-day return in percentage points. $\text{LLM}_{i,t}$ is the Llama-3.3-70B headline-direction signal in $\{-1,0,+1\}$. inner_confidence is the joint label probability of the chosen answer.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$r_{i,t+1}$ (%)	0.061	4.302	−2.599	−1.036	0.035	1.136	2.626	91,361
$\text{LLM}_{i,t}$	0.084	0.937	−1.000	−1.000	0.000	1.000	1.000	91,361
$\text{LAP}^{P(\text{known})}_{i,t+1}$	0.414	0.415	0.001	0.014	0.226	0.922	0.999	91,361
$(U-D)_{i,t+1}$	0.128	0.480	−0.492	0.000	0.024	0.370	0.880	91,361
inner_confidence	0.986	0.059	0.987	0.999	1.000	1.000	1.000	91,361
Salience proxies								
Market cap ($\$M$)	195.2	368.6	3.9	19.7	71.4	201.0	457.0	91,718
σ_{12m}	0.105	0.204	0.045	0.058	0.079	0.110	0.163	90,647
News volume	10.573	22.415	0.000	0.000	2.000	9.000	33.000	92,003
Analyst coverage	1.784	7.531	0.000	0.000	0.000	0.000	0.000	92,003
Trading volume ($\times 10^6$)	19.632	45.154	0.717	1.860	5.301	17.829	50.541	92,003

Panel B: Out-of-sample

This panel presents summary statistics for the variables used in the stock-news analysis on the out-of-sample window (year 2024). Variables are defined as in Panel A.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$r_{i,t+1}$ (%)	0.095	4.729	−2.934	−1.114	0.066	1.227	2.837	7,568
$\text{LLM}_{i,t}$	0.089	0.940	−1.000	−1.000	0.000	1.000	1.000	7,568
$\text{LAP}^{P(\text{known})}_{i,t+1}$ ($\times 10^{-6}$)	7.250	5.676	2.906	3.749	5.490	8.972	13.119	7,568
$(U-D)_{i,t+1}$ ($\times 10^{-6}$)	7.139	5.537	2.855	3.716	5.413	8.871	12.929	7,568
inner_confidence	0.986	0.059	0.989	0.999	1.000	1.000	1.000	7,568

Table A.4. Summary Statistics: Earnings-Call Capital-Expenditure Prediction

Panel A: In-sample

This panel presents summary statistics for the variables used in the earnings-call corporate-investment analysis on the in-sample window (2006Q1–2020Q4). $\text{CapEx}_{i,q+2}$ is the firm's capital-expenditure-to-assets ratio realized two quarters ahead, in percent, winsorized at the 1%/99% level within yq. $\text{LAP}^{P(\text{known})}_{i,q+2}$ is the lookahead propensity from the date-only capex-direction recall query at quarter $q+2$. $\text{LLM}_{i,q}$ is the Llama-3.3-70B capex-direction signal from the truncated earnings-call transcript ($\{-1, -0.5, 0, +0.5, +1\}$).

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\text{CapEx}_{i,q+2}$ (%)	2.589	3.583	0.132	0.505	1.378	3.174	6.290	116,746
$\text{LLM}_{i,q}$	0.174	0.455	−0.500	0.000	0.500	0.500	0.500	116,746
$\text{LAP}^{P(\text{known})}_{i,q+2}$	0.027	0.124	0.000	0.000	0.000	0.001	0.016	116,746
$(U-D)_{i,q+2}$	0.005	0.116	−0.001	−0.000	−0.000	0.000	0.002	116,746
inner_confidence	0.911	0.139	0.677	0.883	0.983	0.998	0.999	116,746
Salience proxies								
Market cap ($\$M$)	8,765	34,551	172	455	1,408	4,664	16,219	107,237
σ_{12m}	0.094	0.081	0.022	0.045	0.077	0.122	0.178	81,124
Analyst coverage	1.997	5.635	0.000	0.000	0.000	0.000	8.000	107,237

Panel B: Out-of-sample

This panel presents summary statistics for the variables used in the earnings-call corporate-investment analysis on the out-of-sample window (2023Q3–2024Q1). Variables are defined as in Panel A.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\text{CapEx}_{i,q+2}$ (%)	1.602	2.159	0.031	0.230	0.817	2.050	4.211	6,744
$\text{LLM}_{i,q}$	0.202	0.448	−0.500	0.000	0.500	0.500	0.500	6,744
$\text{LAP}^{P(\text{known})}_{i,q+2}$ ($\times 10^{-7}$)	1.926	3.281	0.346	0.603	1.071	1.990	3.816	6,744
$(U-D)_{i,q+2}$ ($\times 10^{-7}$)	0.989	2.902	−0.098	0.000	0.264	0.847	2.311	6,744
inner_confidence	0.920	0.131	0.706	0.905	0.986	0.998	0.999	6,744

Table A.5. Summary Statistics: CPI Inflation Prediction

Panel A: In-sample

This panel presents summary statistics for the CPI inflation analysis on the in-sample window (January 1970 to December 2023, 647 months). $\Delta\pi_t^{\text{YoY}}$ is the month-over-month change in U.S. year-over-year CPI inflation, in percentage points. LLM_t is the Llama-3.3-70B directional forecast signal. $\text{LAP}_{t+1}^{P(\text{known})}$ is the lookahead propensity from the date-only recall query asking whether the CPI inflation change went up or down. π_t^{YoY} is the level of year-over-year CPI inflation.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\Delta\pi_t^{\text{YoY}}$ (pp)	−0.004	0.408	−0.336	−0.154	−0.008	0.136	0.366	647
LLM_t	0.164	0.189	−0.200	0.200	0.200	0.200	0.200	647
$\text{LAP}_{t+1}^{P(\text{known})}$	0.999	0.029	0.998	1.000	1.000	1.000	1.000	647

Panel B: Out-of-sample

This panel presents summary statistics for the CPI inflation analysis on the out-of-sample window (January 2024 to July 2025, 19 months). Variables are defined as in Panel A.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\Delta\pi_t^{\text{YoY}}$ (pp)	−0.031	0.203	−0.285	−0.161	−0.032	0.068	0.197	19
LLM_t	0.032	0.203	−0.200	−0.200	0.200	0.200	0.200	19
$\text{LAP}_{t+1}^{P(\text{known})}$ ($\times 10^{-3}$)	0.699	3.204	0.000	0.000	0.001	0.002	0.018	19

Table A.6. Summary Statistics: Chinese Stock Market (CSI 300)

Panel A: In-sample

This panel presents summary statistics for the Chinese stock market analysis on the in-sample window (January 2021 to December 2023, 427 CSI 300 constituents). $\text{LAP}_{i,t+1}^{P(\text{known})}$ is the lookahead propensity from the Llama-3.3-70B date-only recall query for the next trading day. $(U-D)_{i,t+1}$ is the directional signal $P(\text{up})-P(\text{down})$.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\text{LAP}_{i,t+1}^{P(\text{known})}$ ($\times 10^{-5}$)	4.308	24.594	0.290	0.542	1.474	3.536	7.485	276,103
$(U-D)_{i,t+1}$ ($\times 10^{-5}$)	4.307	24.595	0.290	0.542	1.474	3.535	7.484	276,103

Panel B: Out-of-sample

This panel presents summary statistics for the Chinese stock market analysis on the out-of-sample window (January 2024 to December 2025, 425 CSI 300 constituents). Variables are defined as in Panel A.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\text{LAP}_{i,t+1}^{P(\text{known})}$ ($\times 10^{-6}$)	1.581	0.982	0.735	0.962	1.372	1.995	2.603	196,829
$(U-D)_{i,t+1}$ ($\times 10^{-6}$)	1.553	0.976	0.714	0.941	1.349	1.948	2.560	196,829

Table A.7. Two-Sample t-Tests: In-Sample vs. Post-Cutoff $\mathrm{LAP}^{P(\mathrm{known})}$

This table reports two-sample t-tests (Welch's) comparing the in-sample and post-cutoff means of $\mathrm{LAP}^{P(\mathrm{known})}$ for each application.

Application	IS mean	OOS mean	Difference	t-statistic	Regime
News Headlines	0.412	≈ 0	0.412	301.4	Variation
Earnings Calls	0.027	≈ 0	0.027	73.4	Near zero
CPI Inflation	0.999	0.001	0.998	749.4	Saturation
CSI 300	0.00004	0.000002	0.00004	88.7	Zero exposure

Table A.8. Summary Statistics: GPT-4o-mini Stock News Prediction

Panel A: In-sample

This panel presents summary statistics for the GPT-4o-mini variables on the in-sample window (January 2012 to October 2023). GPT-4o-mini has a knowledge cutoff of October 2023. $\text{LLM}_{i,t}$ is the GPT-4o-mini headline-direction signal in $\{-1, 0, +1\}$. $\text{LAP}^{P(\text{known})}_{i,t+1}$ is the lookahead propensity from the GPT-4o-mini date-only recall query. $(U-D)_{i,t+1}$ is the recall-query directional signal.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\text{LLM}_{i,t}$	0.046	0.913	−1.000	−1.000	0.000	1.000	1.000	90,253
$\text{LAP}^{P(\text{known})}_{i,t+1}$	0.169	0.318	0.000	0.000	0.004	0.110	0.864	90,253
$(U-D)_{i,t+1}$	0.010	0.316	−0.156	−0.007	0.000	0.000	0.207	90,253

Panel B: Out-of-sample

This panel presents summary statistics on the out-of-sample window (November 2023 onward). Variables are defined as in Panel A.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\text{LLM}_{i,t}$	0.061	0.930	−1.000	−1.000	0.000	1.000	1.000	7,103
$\text{LAP}^{P(\text{known})}_{i,t+1}$ ($\times 10^{-8}$)	3.028	102.445	0.000	0.000	0.000	0.004	0.000	7,103
$(U-D)_{i,t+1}$ ($\times 10^{-8}$)	−1.616	85.479	0.000	0.000	0.000	0.000	0.000	7,103

Table A.9. Summary Statistics: GPT-4o-mini Earnings-Call Capital-Expenditure Prediction

Panel A: In-sample

This panel presents summary statistics for the GPT-4o-mini variables on the in-sample window (recall quarters through October 2023). $\text{LLM}_{i,q}$ is the GPT-4o-mini capex-direction signal ($\{-1, -0.5, 0, +0.5, +1\}$). $\text{LAP}^{P(\text{known})}_{i,q+2}$ is the lookahead propensity from the GPT-4o-mini date-only recall query at quarter $q+2$. $(U-D)_{i,q+2}$ is the recall-query directional signal.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\text{LLM}_{i,q}$	0.328	0.348	−0.500	0.000	0.500	0.500	0.500	116,531
$\text{LAP}^{P(\text{known})}_{i,q+2}$	0.082	0.214	0.000	0.000	0.000	0.022	0.269	116,531
$(U-D)_{i,q+2}$	0.014	0.203	−0.023	−0.001	0.000	0.000	0.029	116,531

Panel B: Out-of-sample

This panel presents summary statistics on the out-of-sample window (recall quarters after October 2023). Variables are defined as in Panel A.

Variable	Mean	SD	P10	P25	Median	P75	P90	N
$\text{LLM}_{i,q}$	0.359	0.340	−0.500	0.000	0.500	0.500	0.500	13,325
$\text{LAP}^{P(\text{known})}_{i,q+2}$ $(\times 10^{-8})$	2.699	244.876	0.000	0.000	0.000	0.000	0.000	13,325
$(U-D)_{i,q+2}$ $(\times 10^{-8})$	0.399	32.935	0.000	0.000	0.000	0.000	0.000	13,325

Table A.10. Conditional Slope of LLM Forecast by LAP Tercile

This table reports the LLM forecast slope estimated within LAP terciles from a single regression: $Y = \sum_{k=1}^{3} \delta_k \cdot (\text{LLM} \times \mathbf{1}[\text{LAP} \in T_k]) + \sum_{k=1}^{3} \gamma_k \cdot \mathbf{1}[\text{LAP} \in T_k] + \text{firm FE} + \text{time FE} + \varepsilon$. Column (1) reports headline results (firm and date FE, clustered by date); Column (2) reports earnings results (firm and quarter FE, clustered by firm). t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level.

	(1) News Headlines $r_{i,t+1}$	(2) Earnings Calls $\text{CapEx}_{i,q+2}$
LLM × $\mathbf{1}[T_1]$ (low LAP)	0.171***	0.402***
	(7.16)	(9.07)
LLM × $\mathbf{1}[T_2]$ (mid)	0.151***	0.478***
	(7.07)	(11.08)
LLM × $\mathbf{1}[T_3]$ (high LAP)	0.302***	0.754***
	(8.19)	(12.86)
Firm FE	Yes	Yes
Date / Quarter FE	Yes	Yes
N	91,357	106,994